SMC of python ABMs

An integration of MultiVeStA and Mesa

Andrea Vandin

Sant'Anna School of Advanced Studies – Pisa

Institute of Economics

Department of Excellence 2018 - 2022

EMbeDS Economics and Management in the era of Data Science

Department of Excellence 2023 - 2027

LEMbeDS Economics, Management and Law in the era of Data Science

DTU Danmarks Tekniske Universitet

Funded by the European Union NextGenerationEU

Ministero dell'Università e della Ricerca

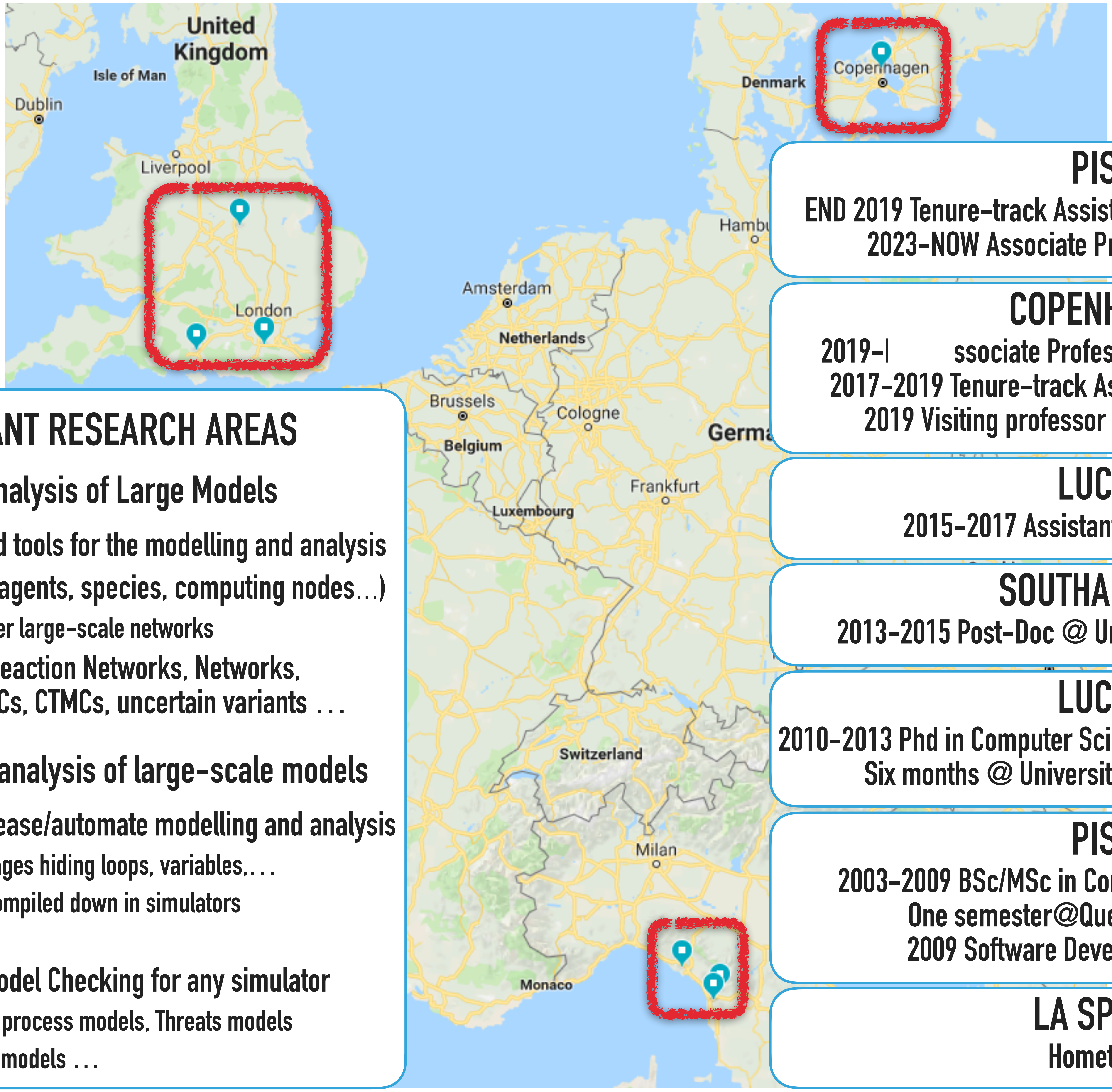
Italiadomani PIANO NAZIONALE DI RIPRESA E RESILIENZA

FAIR Future Artificial Intelligence Research

Outline

- The context at Sant’Anna Pisa
- Economic ABMs
- SMC of Economic ABMs & MultiVeStA
- Two frameworks for ABMs from the social sciences: NetLogo and Mesa
  - Mesa case study: Transient analysis of flocks of Boids
  - Mesa case study: Steady state analysis of Segregation models (Schelling)
- Recent results on graphically explaining SMC results

ABOUT ME



MOST RELEVANT RESEARCH AREAS

Quantitative Analysis of Large Models

- Scalable techniques and tools for the modelling and analysis
- Massive many entities (agents, species, computing nodes…)
  - Possibly interacting over large-scale networks
- ODEs, DAEs, Chemical Reaction Networks, Networks, Boolean networks, DTMCs, CTMCs, uncertain variants …

Simulation/statistical analysis of large-scale models

- High level languages to ease/automate modelling and analysis
  - Domain-specific languages hiding loops, variables,…
  - Models automatically compiled down in simulators
  - One-click analysis
- MultiVeStA: Statistical Model Checking for any simulator
  - Product lines, Business process models, Threats models
  - Agent-based economic models …

PISA

END 2019 Tenure-track Assistant Professor @ Sant'Anna
2023-NOW Associate Professor @ Sant'Anna

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2003-2009 BSc/MSc in Computer Science @ UniPi
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2009 Software Developer @ ION Trading

LA SPEZIA

Hometown

https://www.santannapisa.it/en/andrea-vandin

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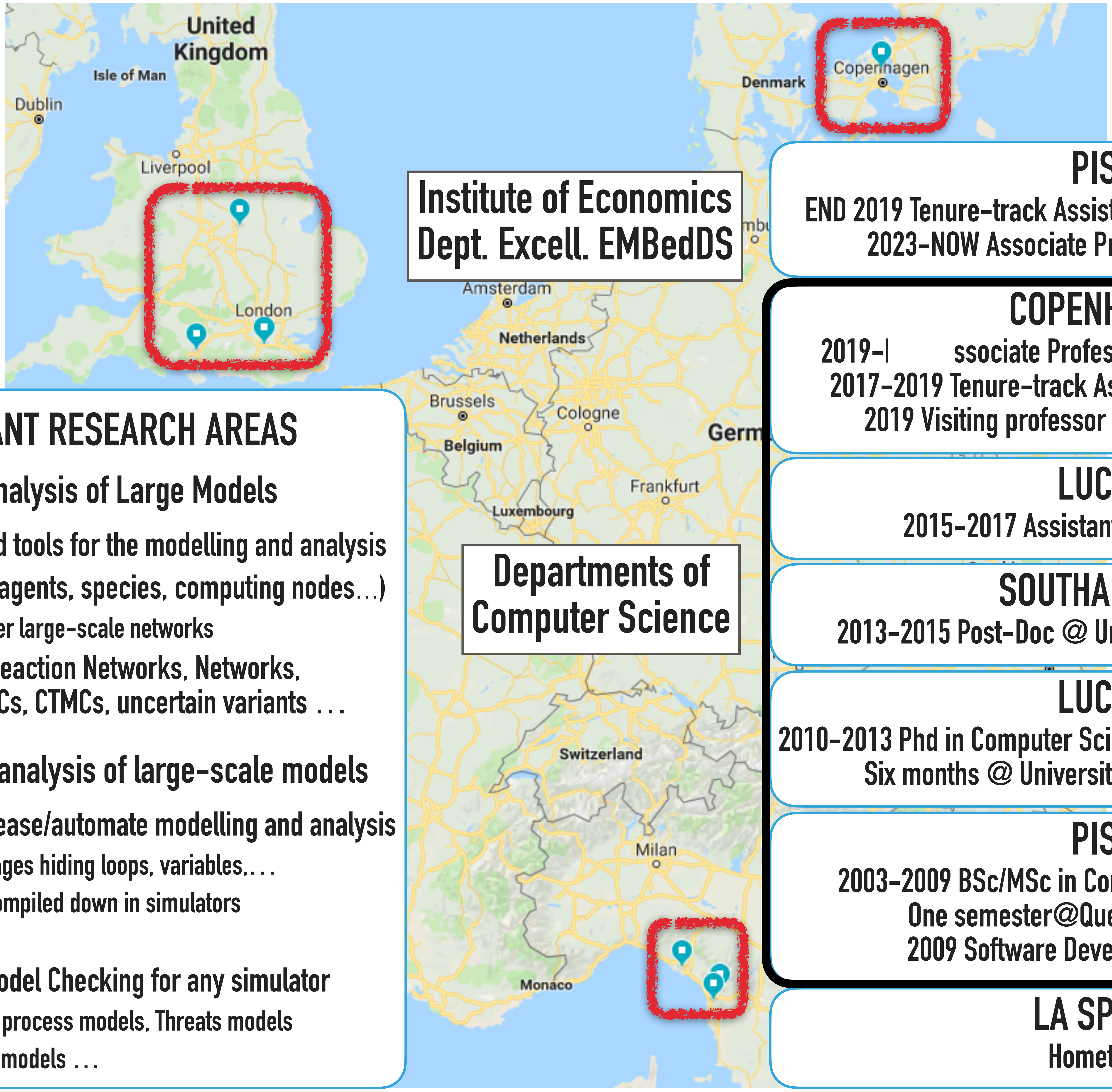

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About L'EMbeDS

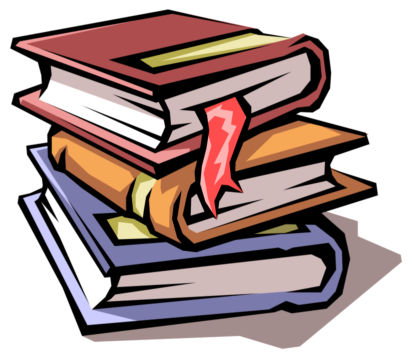
Large funding as 'Department of Excellence' 2018-2022 -> 2023-2027

- Led by Prof. Francesca Chiaromonte
  - Statistician, also working at PennState University, USA
- Involves a wide range of profiles: economics, management, law, statistics, computer science
  - Recently an ACM Paris Kanellakis Theory and Practice Awardee joined: Paolo Ferragina

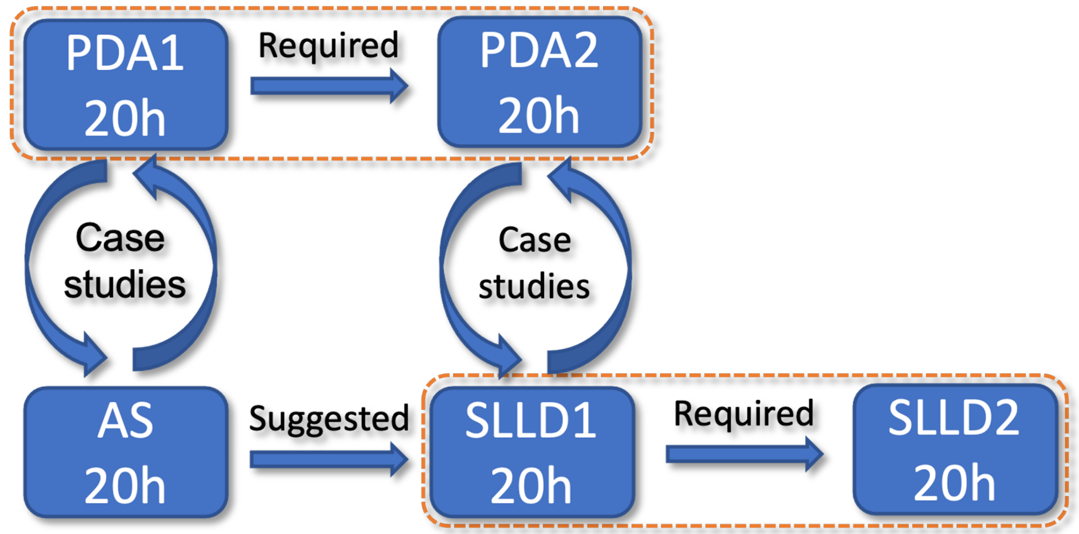
The mission of L'EMbeDS is to

- Foster data-driven statistical and computational approaches in the Social Sciences

Research



Teaching



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About the Institute of Economics

https://www.santannapisa.it/it/istituto/economia/data-science-economics-and-social-science

Research Areas

- Complexity Economics
- The dynamics of industries and markets
- Innovation, technical change and economic history
- Data Science for economics and social science

Data Science for economics and social science

Statistical learning; analysis and methods for big & complex data; data mining; patterns of causality in economic data; statistical model checking; calibration and validation of economic models

An SMC paper in a good Economic Journal

Automated and Distributed Statistical Analysis of Economic Agent-Based Models

Andrea Vandin[a], Daniele Giachini[a], Francesco Lamperti[a,c], Francesca Chiaromonte[a,b]

[a] *Institute of Economics and EMbeDS, Sant'Anna School of Advanced Studies, Pisa, Italy.*
[b] *Dept. of Statistics and Huck Institutes of the Life Sciences, Penn State University, USA*
[c] *RFF-CMCC European Institute on Economics and the Environment, Milan, Italy.*

PUBLISHED AT HIGH-QUALITY ECONOMIC VENUE:
JOURNAL OF ECONOMIC DYNAMICS AND CONTROL 143, 104458, 2022

Abstract

We propose a novel approach to the statistical analysis of simulation models and, especially, agent-based models (ABMs). Our main goal is to provide a fully automated and model-independent tool-kit to inspect simulations and perform counter-factual analysis. Our approach: (i) is easy-to-use by the modeller, (ii) improves reproducibility of results, (iii) optimizes running time given the modeller's machine, (iv) automatically chooses the number of required simulations and simulation steps to reach user-specified statistical confidence, and (v) automatically performs a variety of statistical tests. In particular, our framework is designed to distinguish the transient dynamics of the model from its steady state behaviour (if any), estimate properties of the model in both "phases", and provide indications on the ergodic (or non-ergodic) nature of the simulated processes – which, in turns allows one to gauge the reliability of a steady state analysis. Estimates are equipped with statistical guarantees, allowing for robust comparisons across computational experiments. To demonstrate the effectiveness of our approach, we apply it to two models from the literature: a large scale macro-financial ABM and a small scale prediction market model. Compared to prior analyses of these models, we obtain new insights and we are able to identify and fix some erroneous conclusions.

Keywords: ABM, Statistical Model Checking, Ergodicity analysis, Steady state analysis, Transient analysis, Warmup estimation, T-test and power, Prediction markets, Macro ABM

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What is an Economic Agent-Based Model?

NON ABMs

- 'Mainstream economists' tend to reason in terms of models that
  - Are given as a unique monolithic model
    - Do not have focus on their single components, but on the overall dynamics of the model
    - *What* the system does, rather than *how* the system does
  - Have explicit representations of the laws governing the economic system
  - **Can be analysed analytically**

ABMs

- Some economists are interested in modeling an economy in terms of its components
  - The **agents** that operate in it: firms, households, banks…
- The modeller does NOT specify explicitly the laws governing the model.
- It describes explicitly
  - The behaviour of every agent
  - The interactions among the agents
- **The laws governing the model then *emerge* from these behaviours and interactions**
- These are typically too difficult to be solved analytically
  - We need to do simulations
    - My message: **we need to do simulations well!**
    - Statistical model checking, can help on this

A large Macro ABM

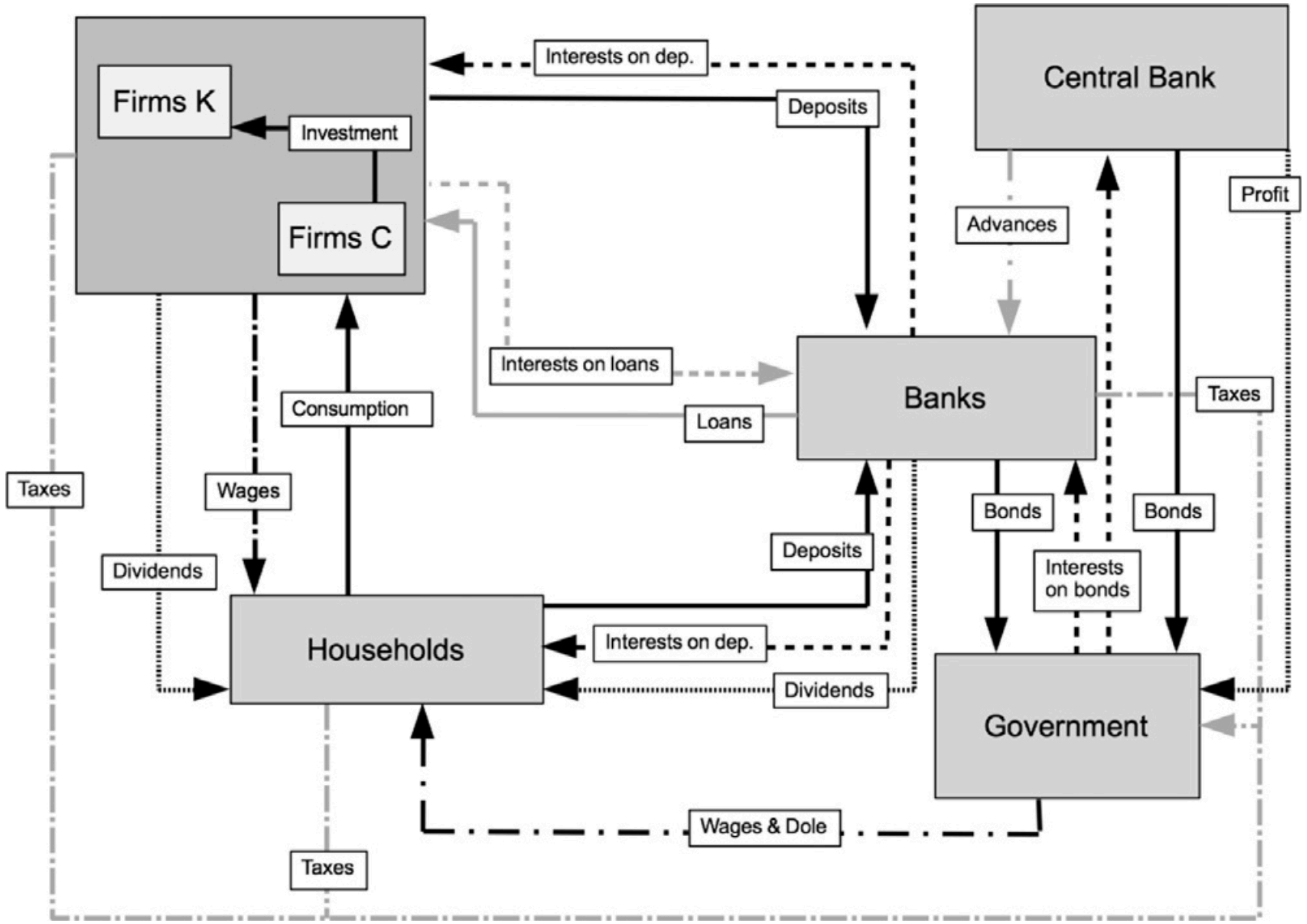

Fig. 1. Flow diagram of the model. Arrows point from paying sectors to receiving sectors.

Large-scale macro financial ABM from JEDC, 2016

- An economy with
  - households,
  - consumption/capital firms,
  - commercial banks
  - government, central bank
- Thousands of agents
- Implemented in JMAB:
  - Java framework for macro ABM models.

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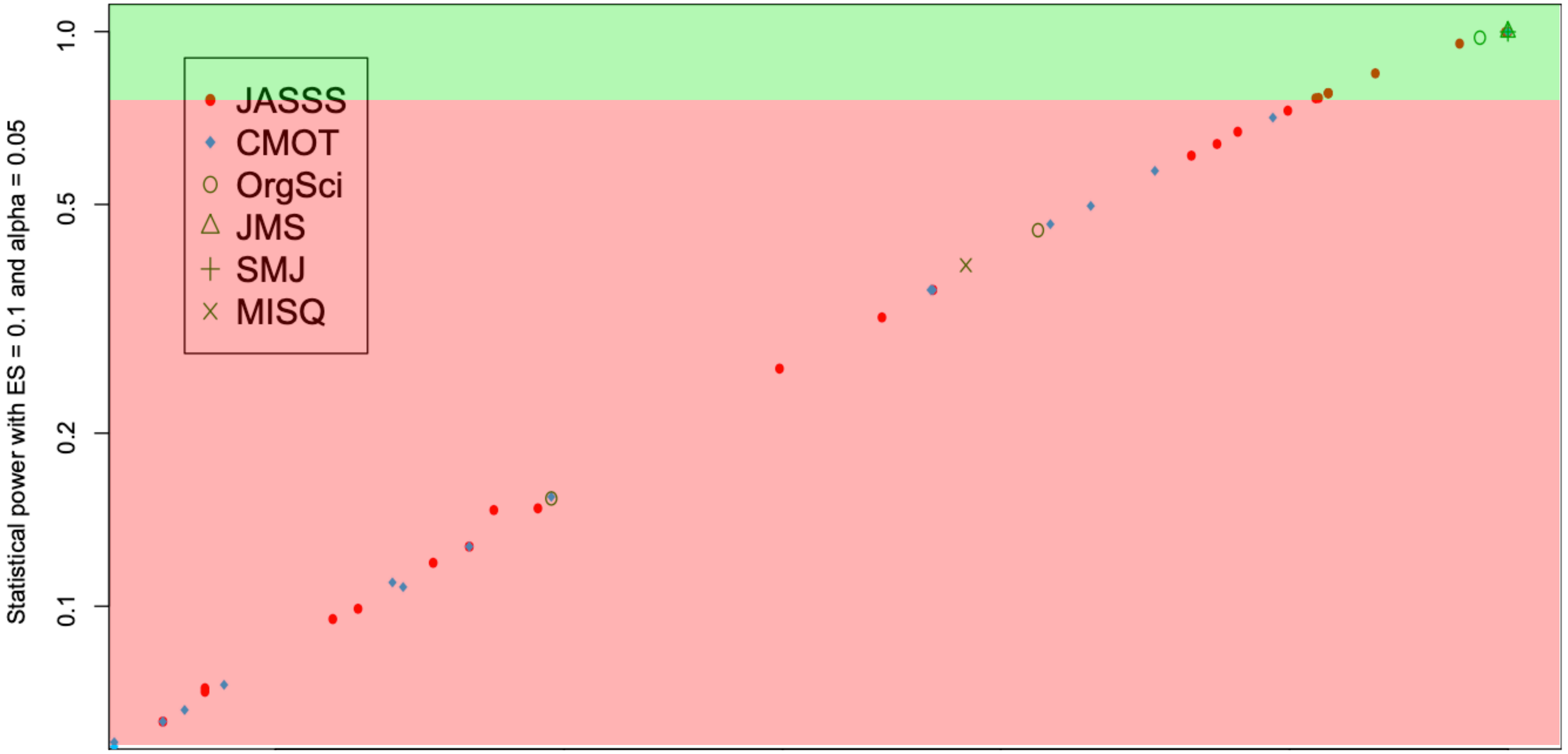
‘Quality’ of Statistical Analysis on 55 ABM

from Management & Organisational Research

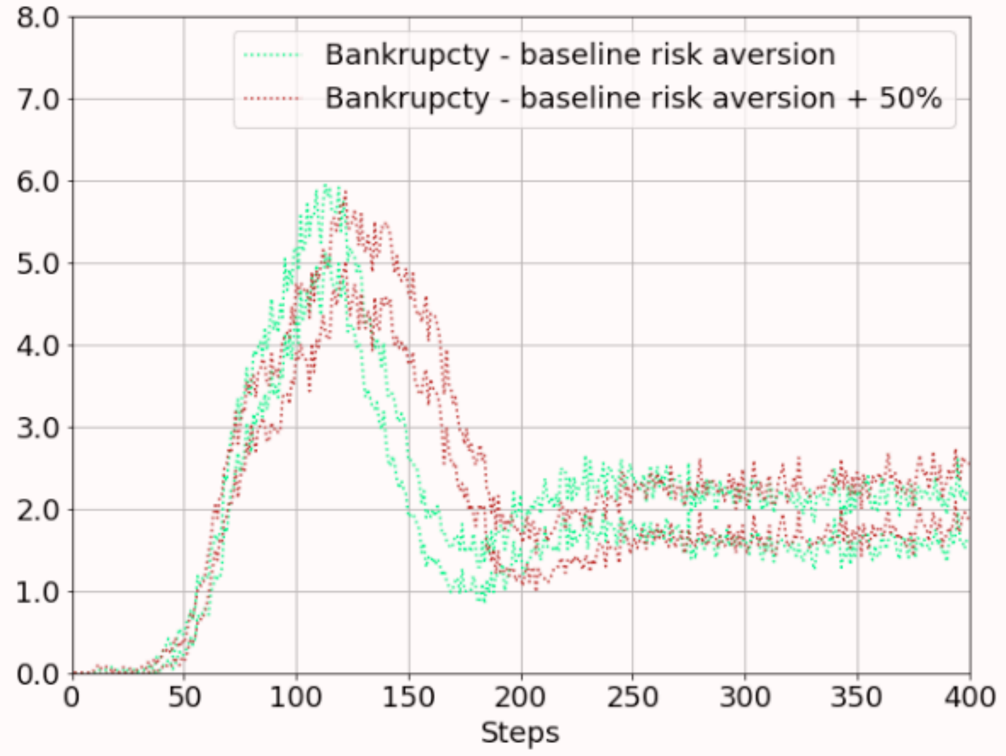


Adapted from Secchi, Seri, Computational and Mathematical Organization Theory, 2017

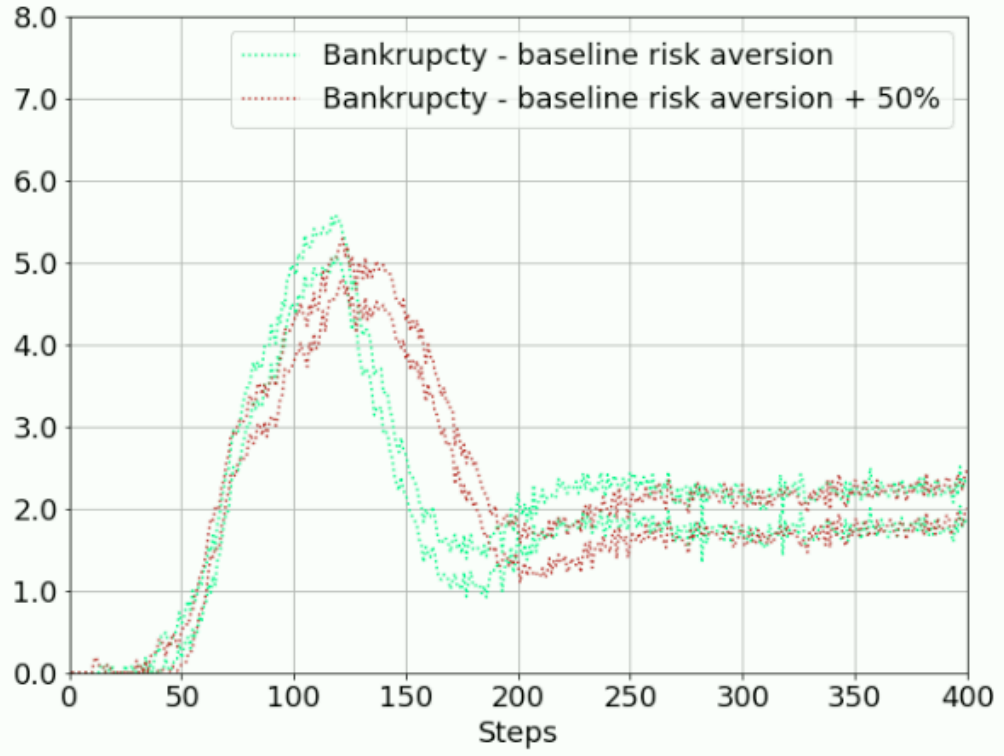
- The importance of designing well simulation-based analysis.
  - Power analysis on ‘are the expected outcomes of different configurations of parameters the same’?
- Power is 1 - P(Type II error)
  - Roughly, P(test=‘outcomes are different’ | outcomes are different)
  - “The value that seems to be more commonly accepted is 80%“
- “We need to encourage researchers to be more precise in the determination of the number of runs”

MultiVeStA: Statistically Meaningful Counterfactual Analysis

97.5% CI
100 Simulations



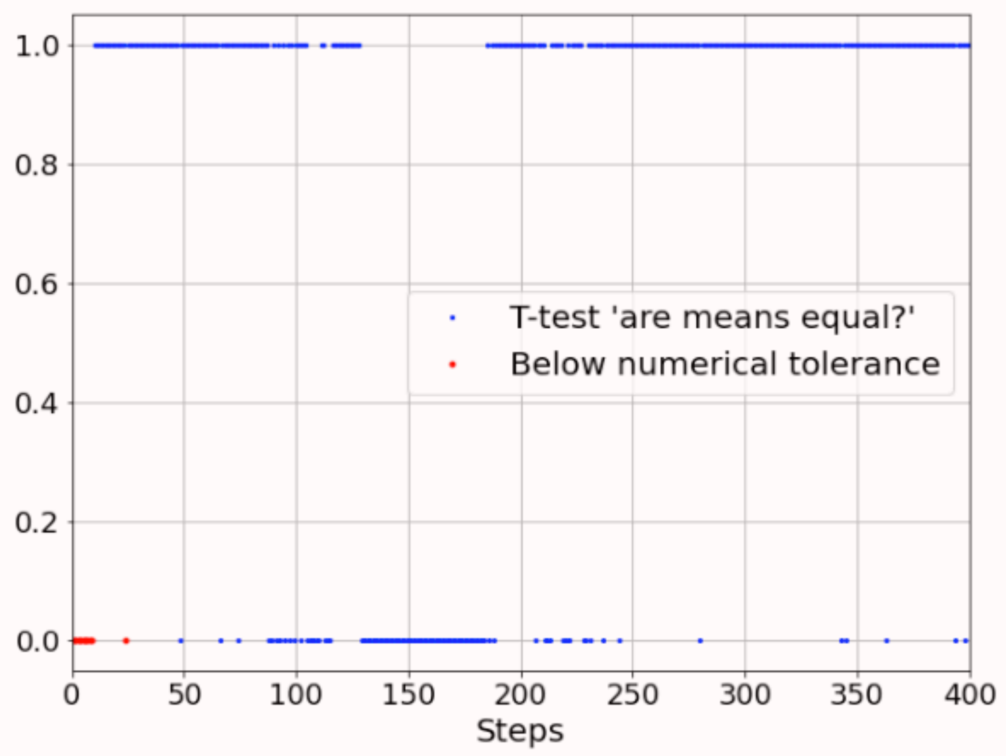
(a) CIs width for $\alpha = 0.025$ and $N = 100$ simulations



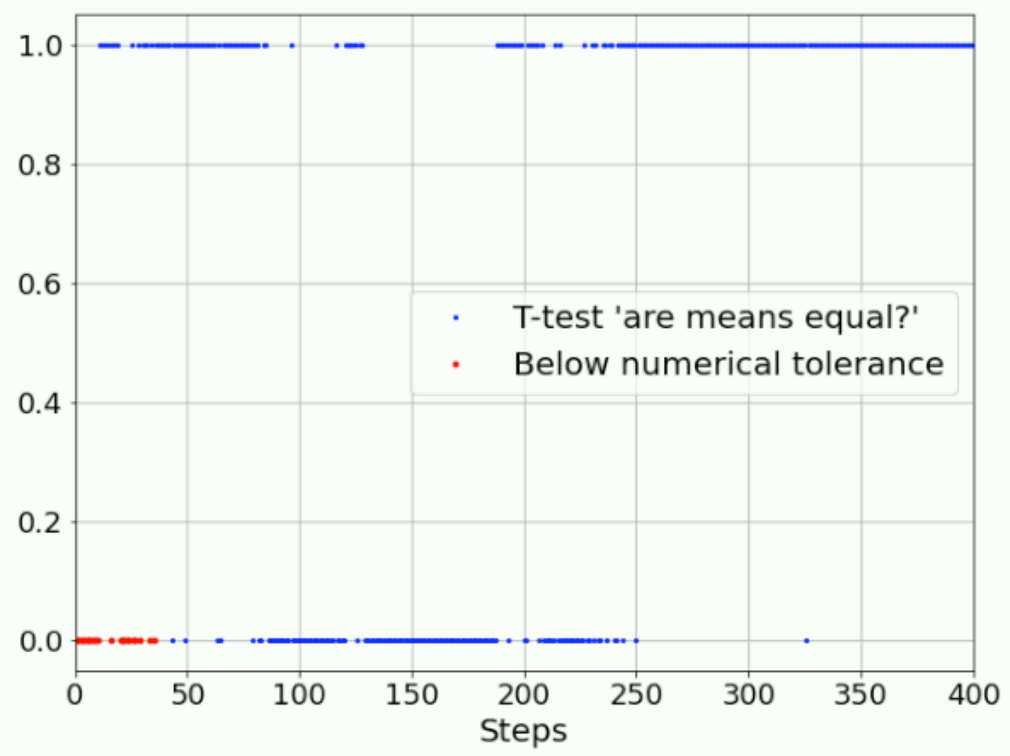
(b) CIs width for $\alpha = 0.025$ and $\delta = 0.5$

97.5% CI
MultiVeStA
'**Right**' number of simulations

Welch's t-test
[Welch1947]



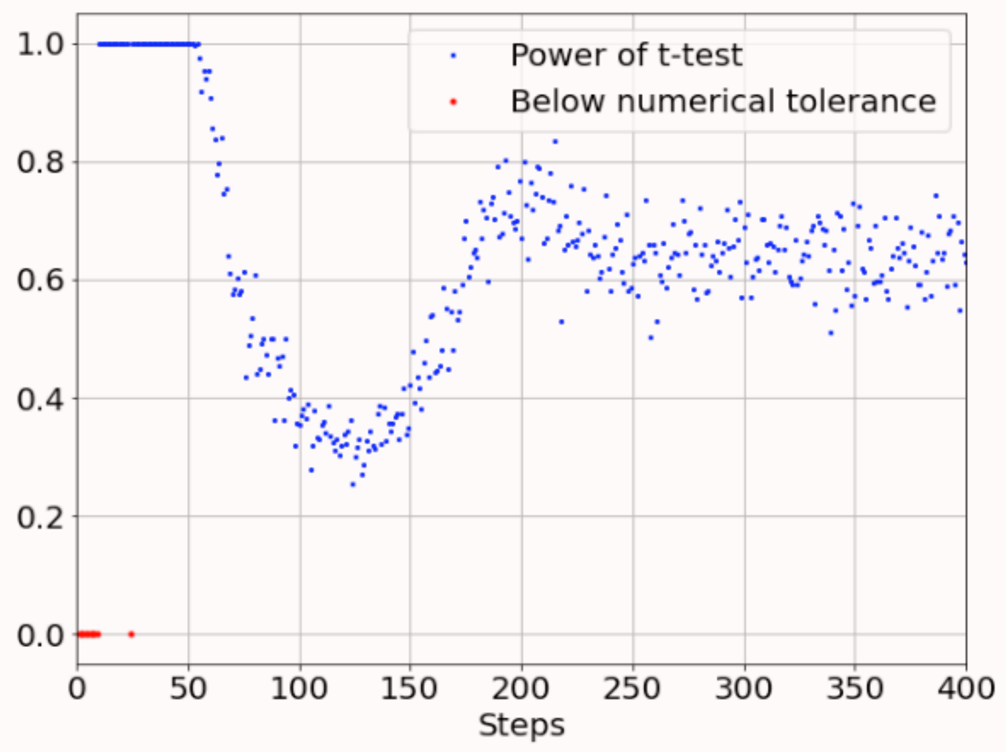
(c) T-test *are means in (a) point-wise equal* for significance $\alpha = 0.025$



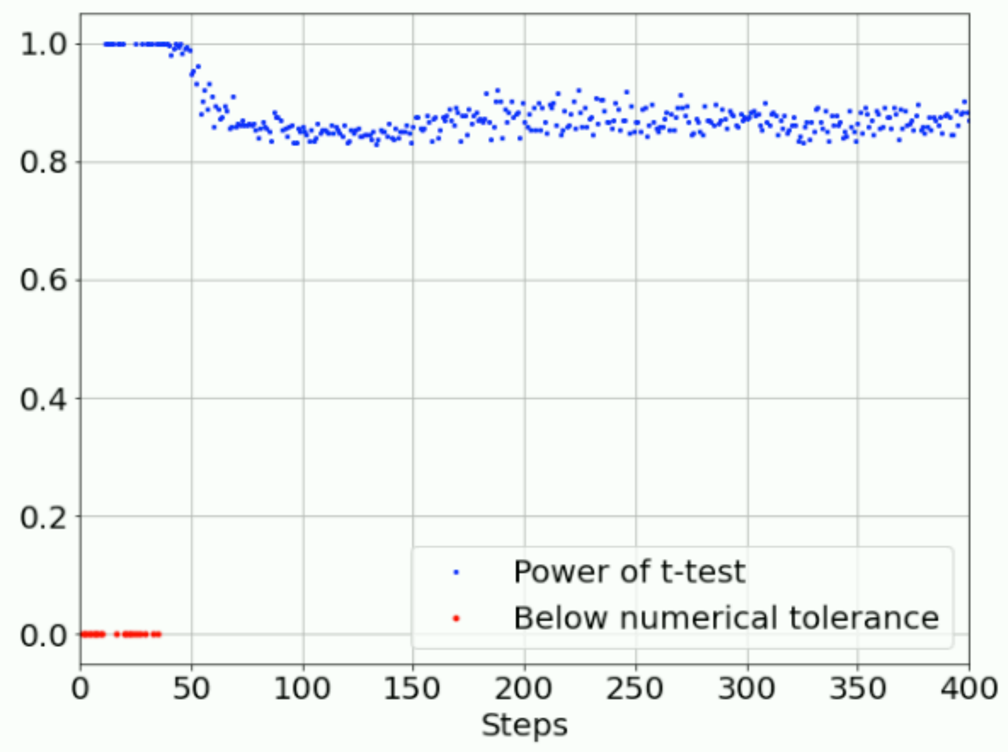
(d) T-test *are means in (b) point-wise equal* for significance $\alpha = 0.025$

Welch's t-test
[Welch1947]

Power of the test
P(Test=0 | Real=0)
[Chow2002]



(e) Power of t-test in (c) for difference $\varepsilon = 0.5$



(f) Power of t-test in (d) for difference $\varepsilon = 0.5$

Power of the test
P(Test=0 | Real=0)
[Chow2002]

MultiVeStA: Steady-State Analysis for Market Selection

Arbitrary choice of
- Number of sims
- Warmup period
- Time horizon

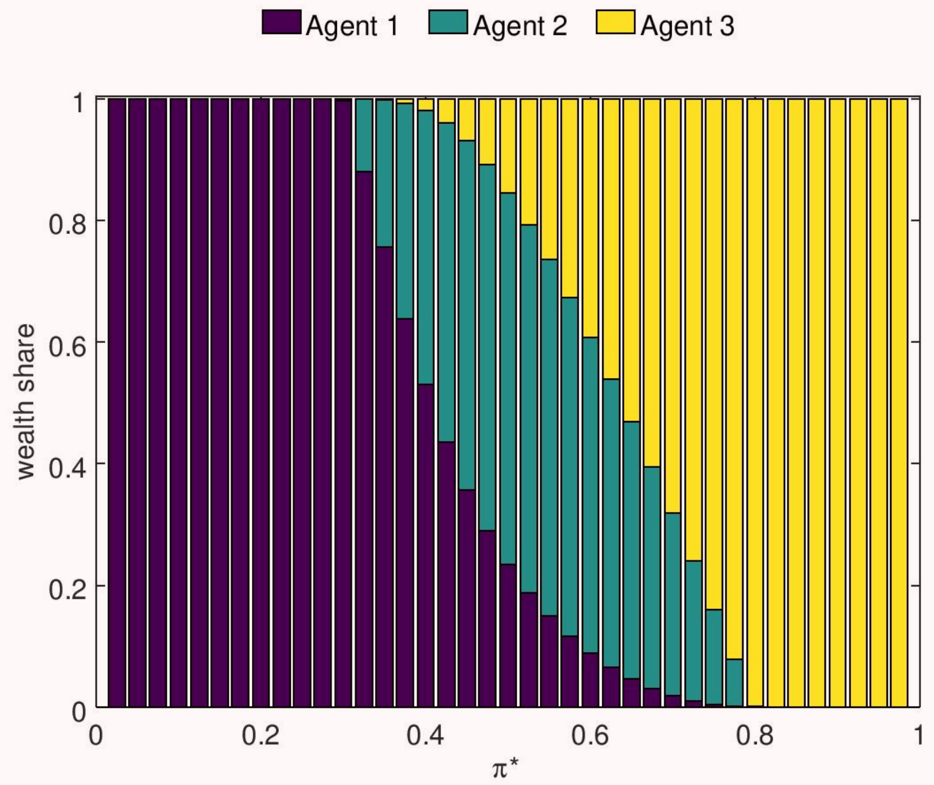
from [Kets et al2014]

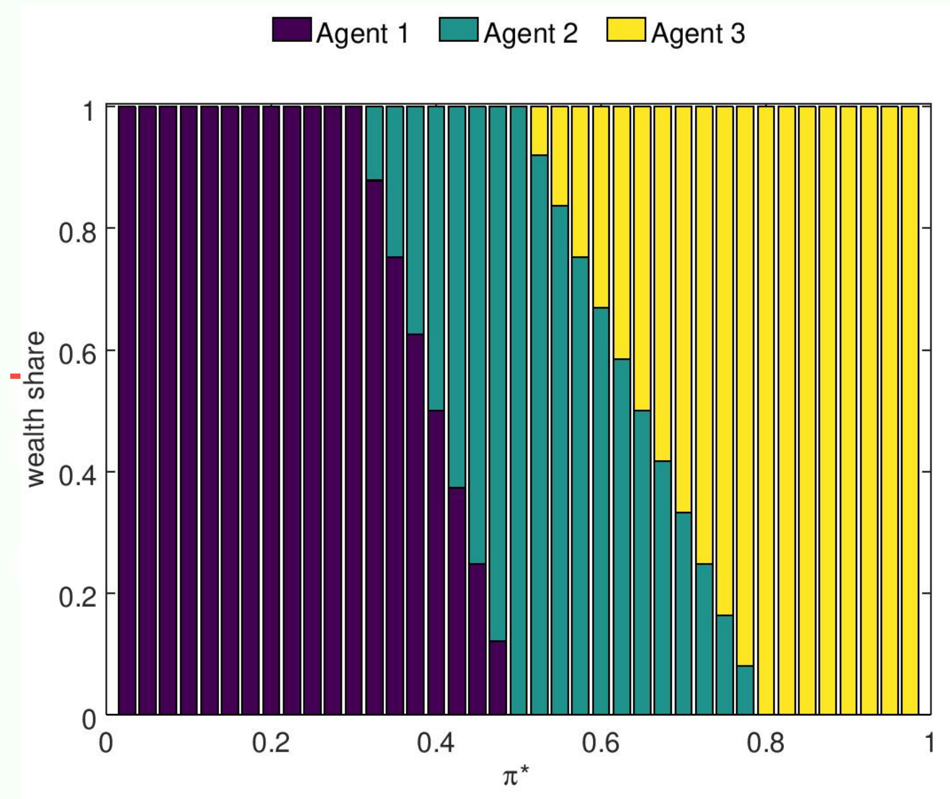
Automated choice of
- Number of sims
- Warmup period
- Time horizon

MultiVeStA

Same as analytical solution
from [Bottazzi,Giachini2019]

Agents wealth at steady state





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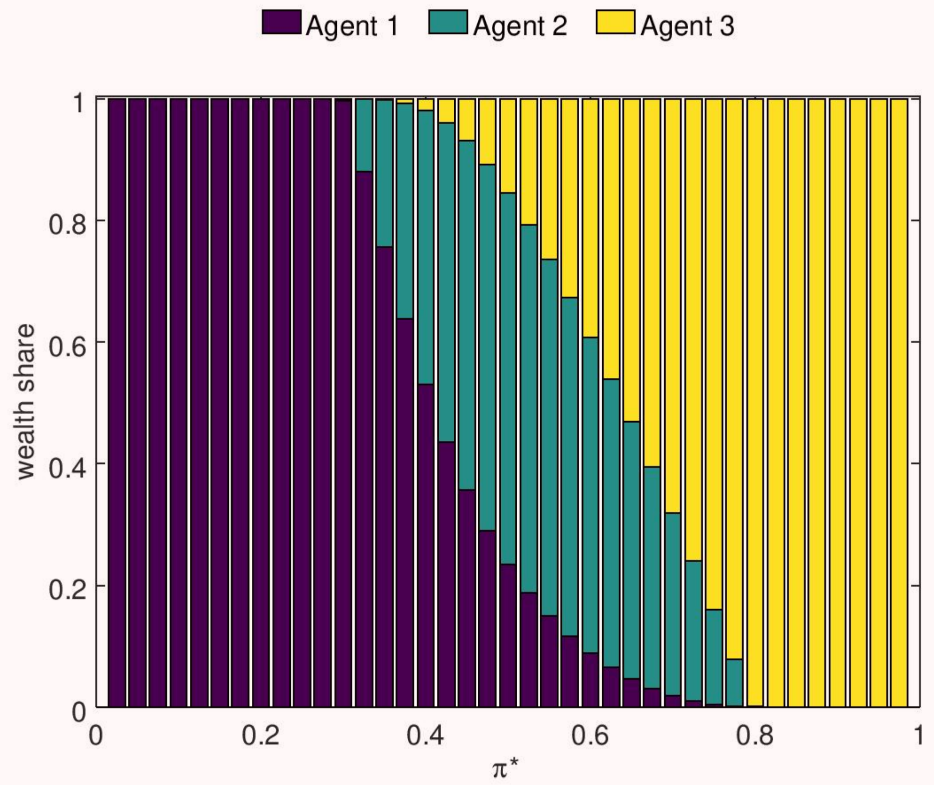
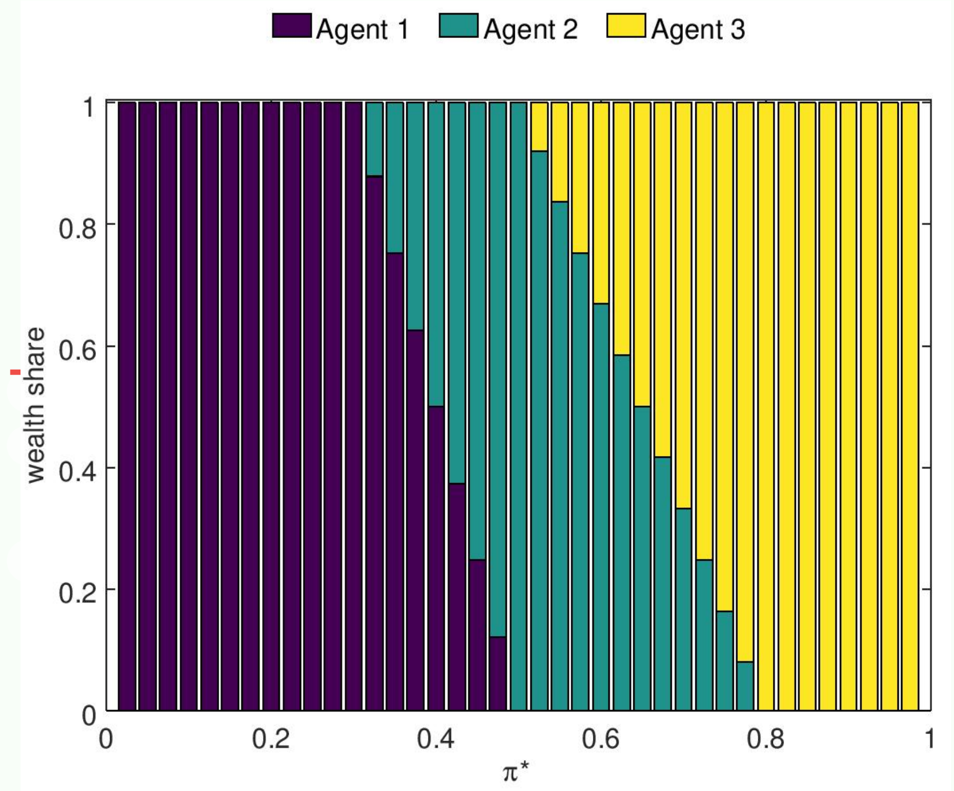
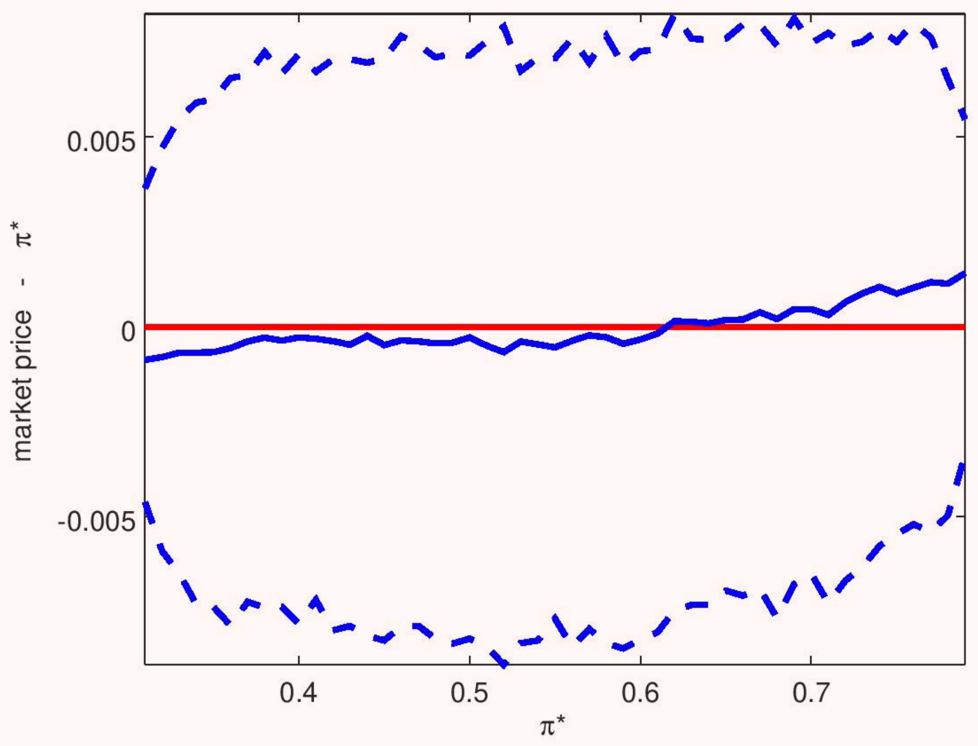
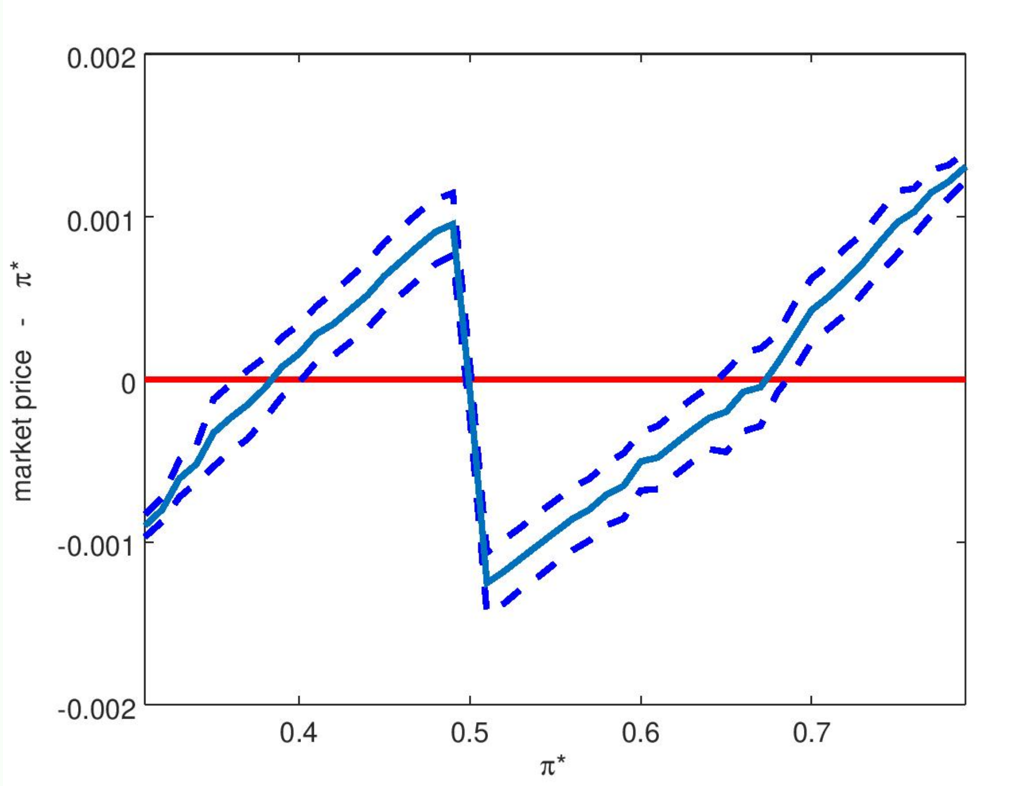




Does the market price match $\pi*$?





MultiVeStA: Transient and Steady-State Algorithms

$$\begin{matrix} y_{1,1} & \boxed{y_{1,2}} & \cdots & y_{1,m} \\ y_{2,1} & \boxed{y_{2,2}} & \cdots & y_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ y_{n,1} & \boxed{y_{n,2}} & \cdots & y_{n,m} \end{matrix}$$

$$\sum_{i=1}^{n} \frac{y_{i,2}}{n} = \overline{Y}_2 \approx E[Y_2]$$

(a) Transient analysis

$$\begin{matrix} y_{1,1} & y_{1,2} & \boxed{\cdots \quad y_{1,m}} \\ y_{2,1} & y_{2,2} & \boxed{\cdots \quad y_{2,m}} \\ \vdots & \vdots & \vdots \quad \vdots \\ y_{n,1} & y_{n,2} & \boxed{\cdots \quad y_{n,m}} \end{matrix}$$

$$\text{(i)} \sum_{t=w+1}^{m} \frac{y_{1,t}}{m-w} = \overline{Y}_1(w)$$

$$\vdots \qquad = \overline{Y}_2(w)$$

$$\vdots \qquad \vdots$$

$$\vdots \qquad = \overline{Y}_n(w)$$

$$\text{(ii)} \sum_{i=1}^{n} \frac{\overline{Y}_i(w)}{n} = \overline{Y}(w) \approx E[Y] = \lim_{t\to\infty} E[Y_t]$$

(b) Steady-state analysis by *Replication and Deletion* (RD)

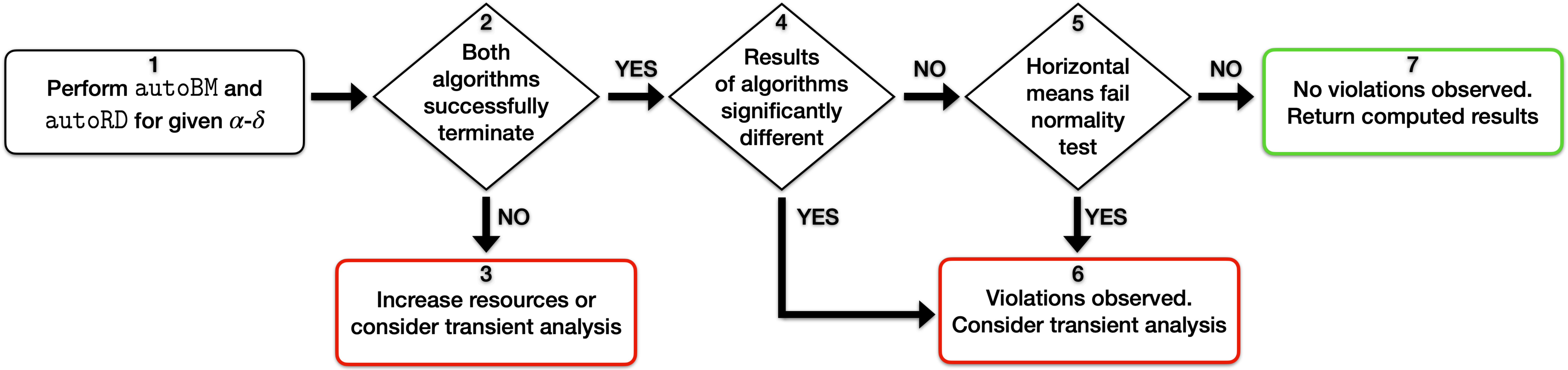
$$\boxed{y_{1,1} \cdots y_{1,b}} \; \boxed{\cdots \; y_{1,2b}} \; \cdots \; \boxed{\cdots \; y_{1,nb}}$$

$$\text{(i)} \sum_{t=1}^{b} \frac{y_{1,t}}{b} = \overline{B}_1 \qquad \overline{B}_2 \qquad \boxed{\cdots \qquad \overline{B}_n}$$

$$\text{(ii)} \sum_{j=l+1}^{n} \frac{\overline{B}_j}{n-l} = \overline{B}(l) \approx E[Y] = \lim_{t\to\infty} E[Y_t]$$

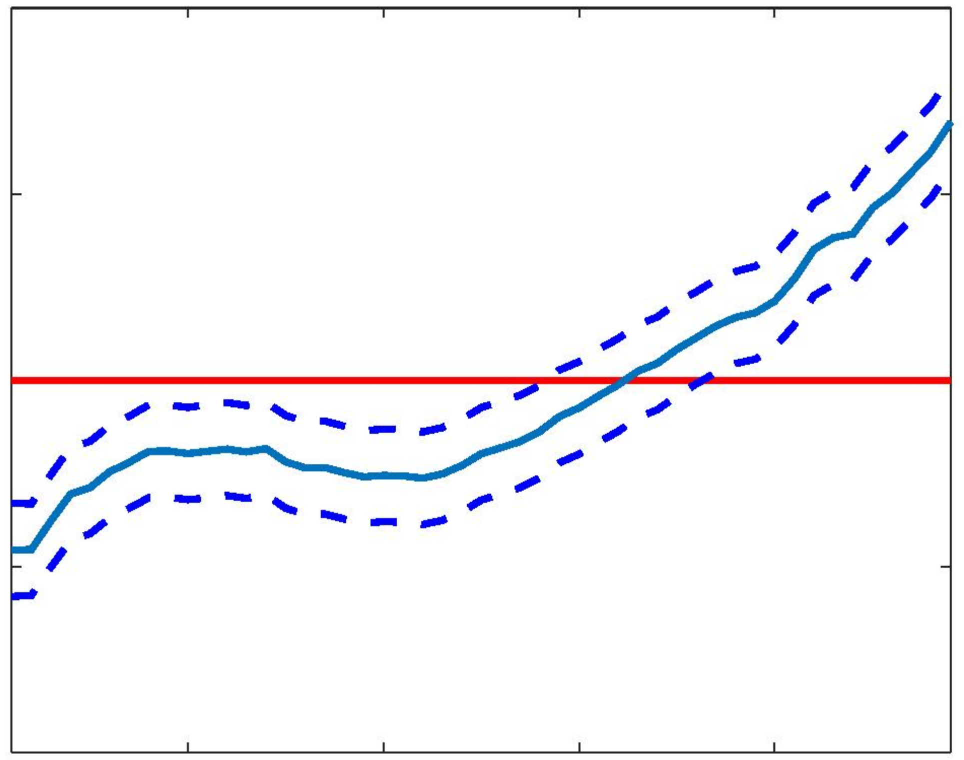
(c) Steady-state analysis by *Batch Means* (BM) using one long simulation

MultiVeStA: a Methodology for Ergodicity Diagnostics



Our Message: our Approach to Simulation-Based Analysis

newstalkzb.co.nz/news/education/modern-lego-sets-more-complex-less-inspiring/



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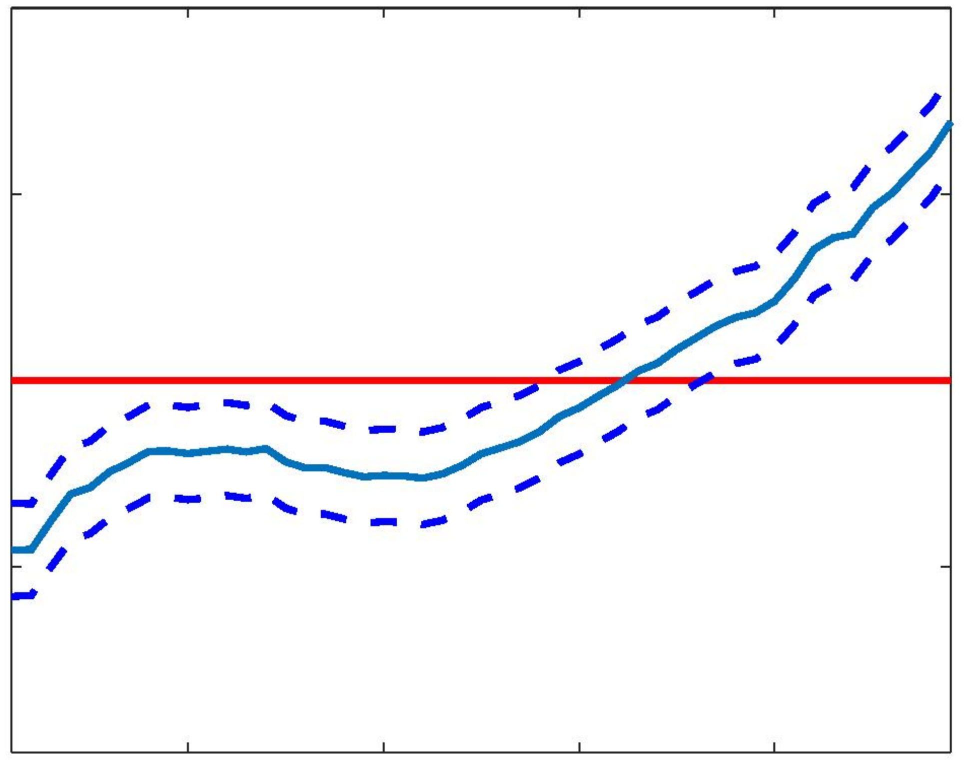
Handcrafted

- Mainly manual process
  - Time-consuming
  - Problems with replicability
  - Error-prone
    - Modify model, interpret CSV
- Ad-hoc implementations
  - Reliability? Efficiency?



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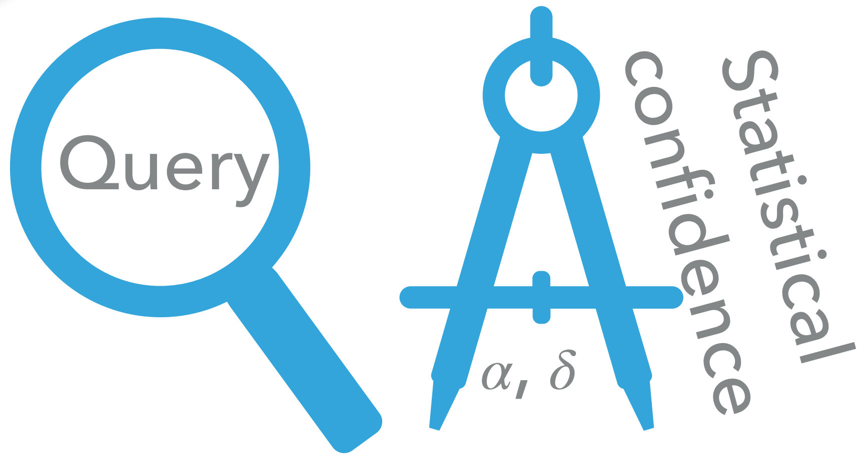

Statistical Model Checking

- Automatic
  - Time-saving and Reproducible
  - Promotes use of *standard* analysis
- Reference implementation
  - Reliable and Efficient

https://www.alamy.com/

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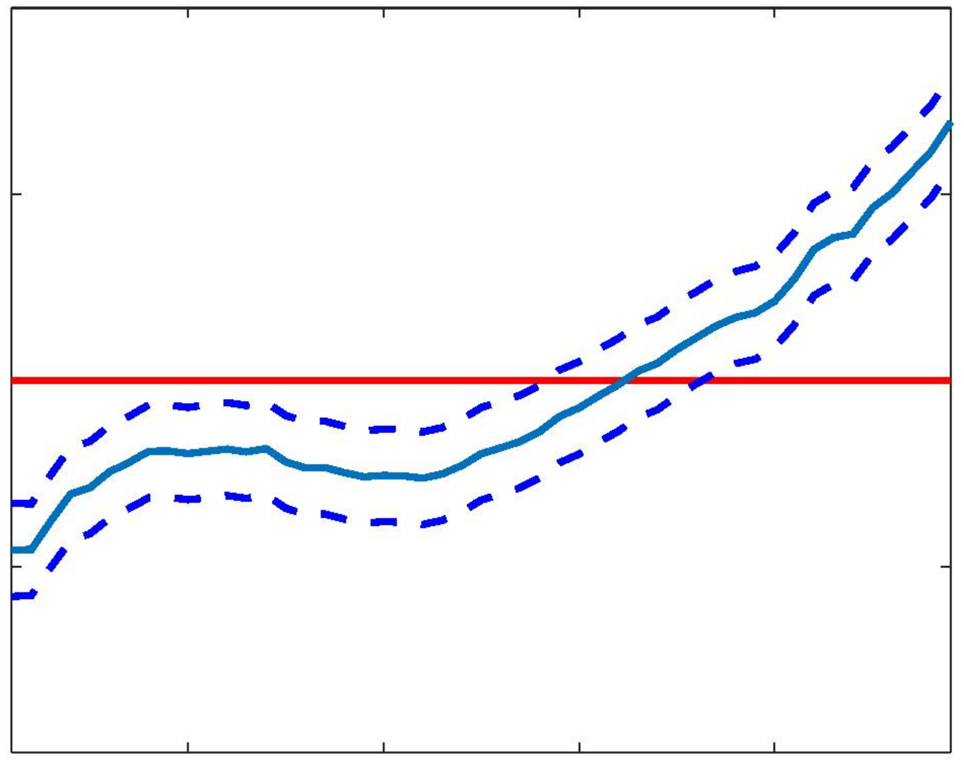

MultiVeStA: SMC For Discrete-Event Simulators

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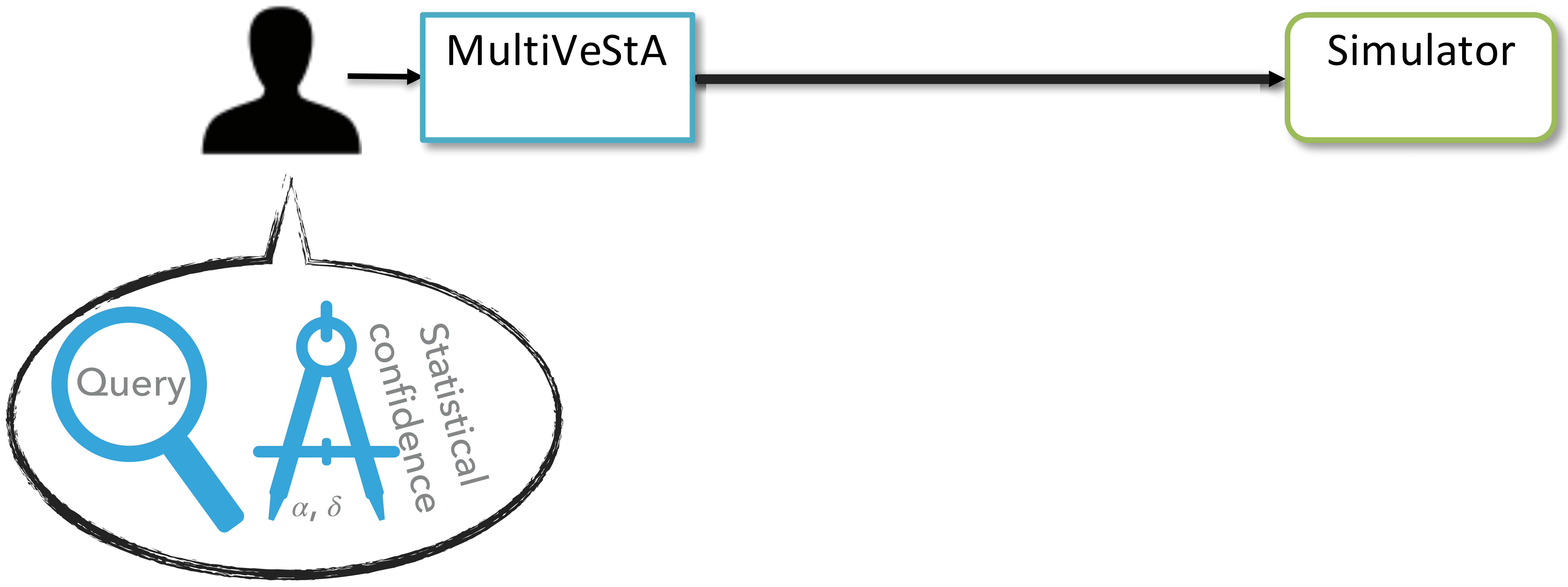

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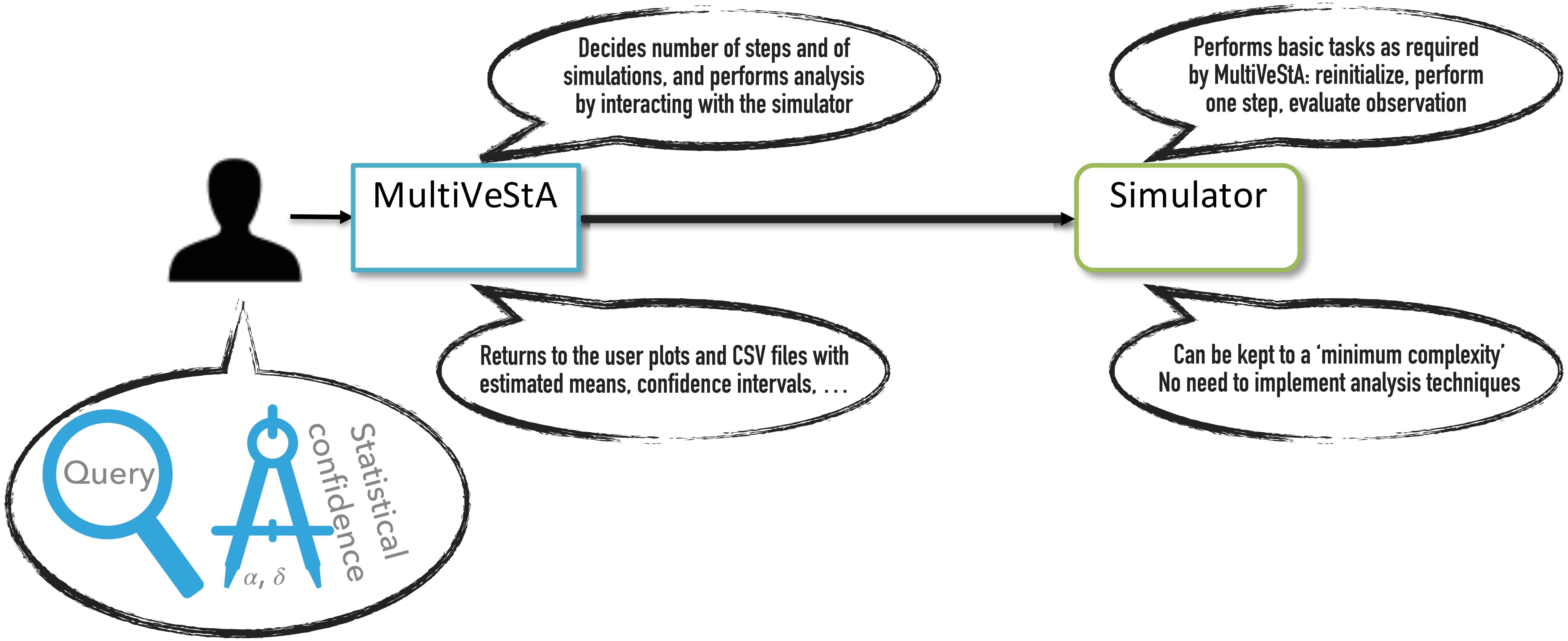

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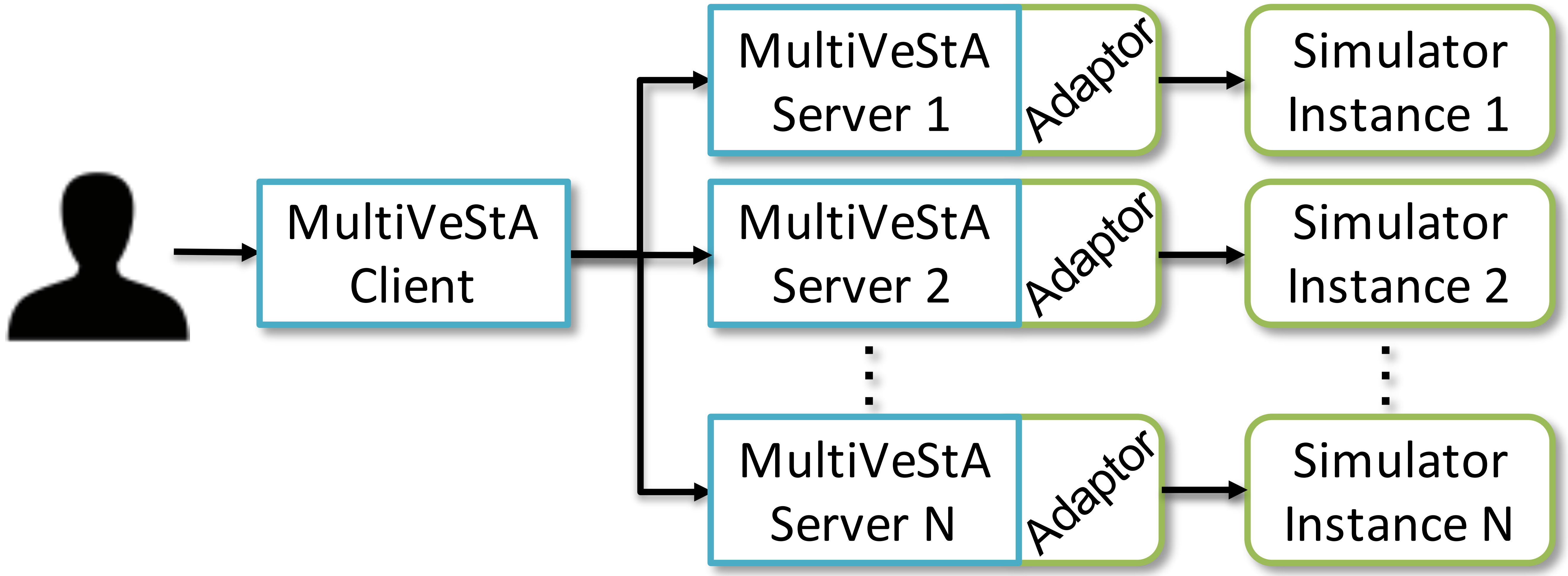

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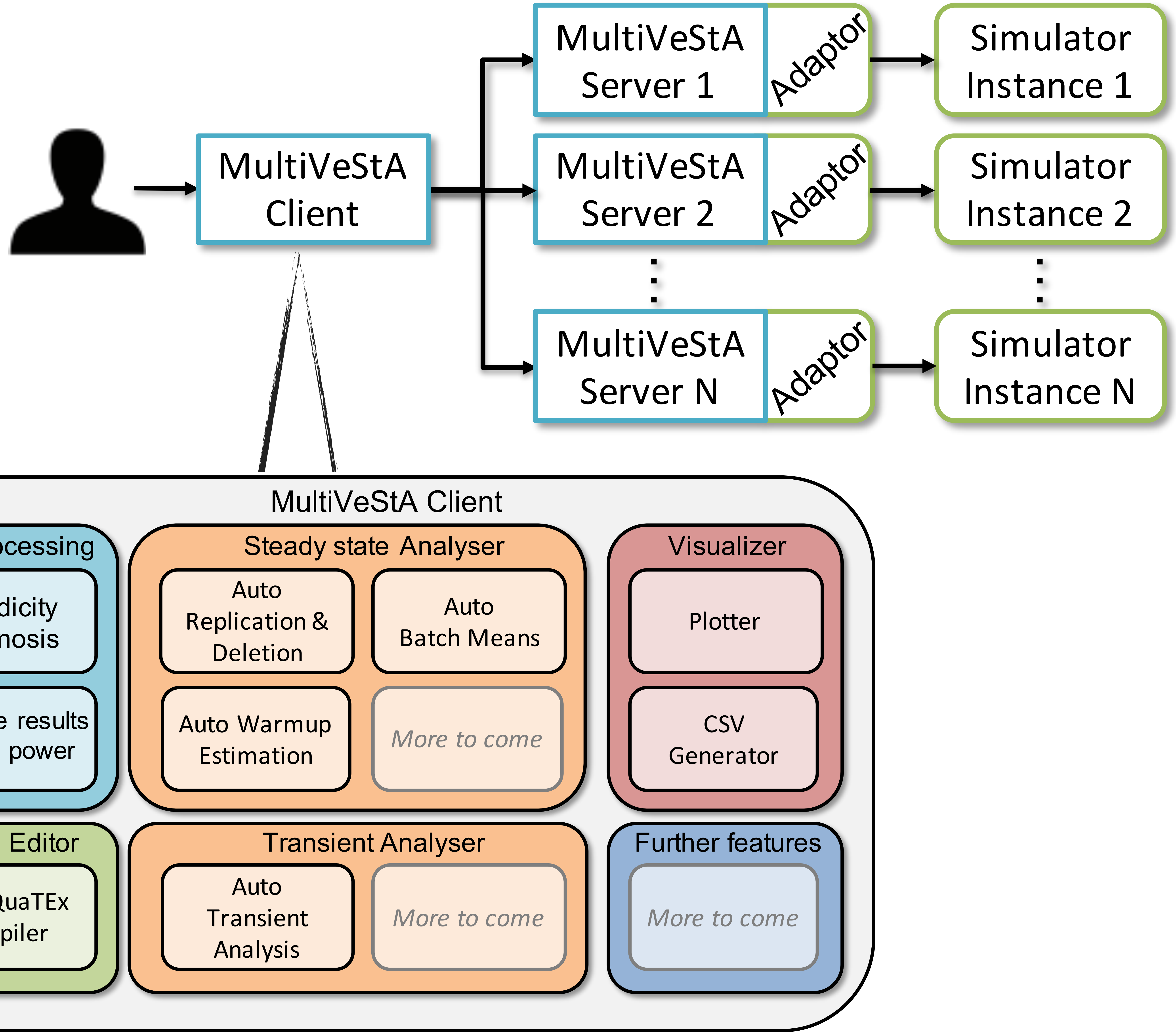

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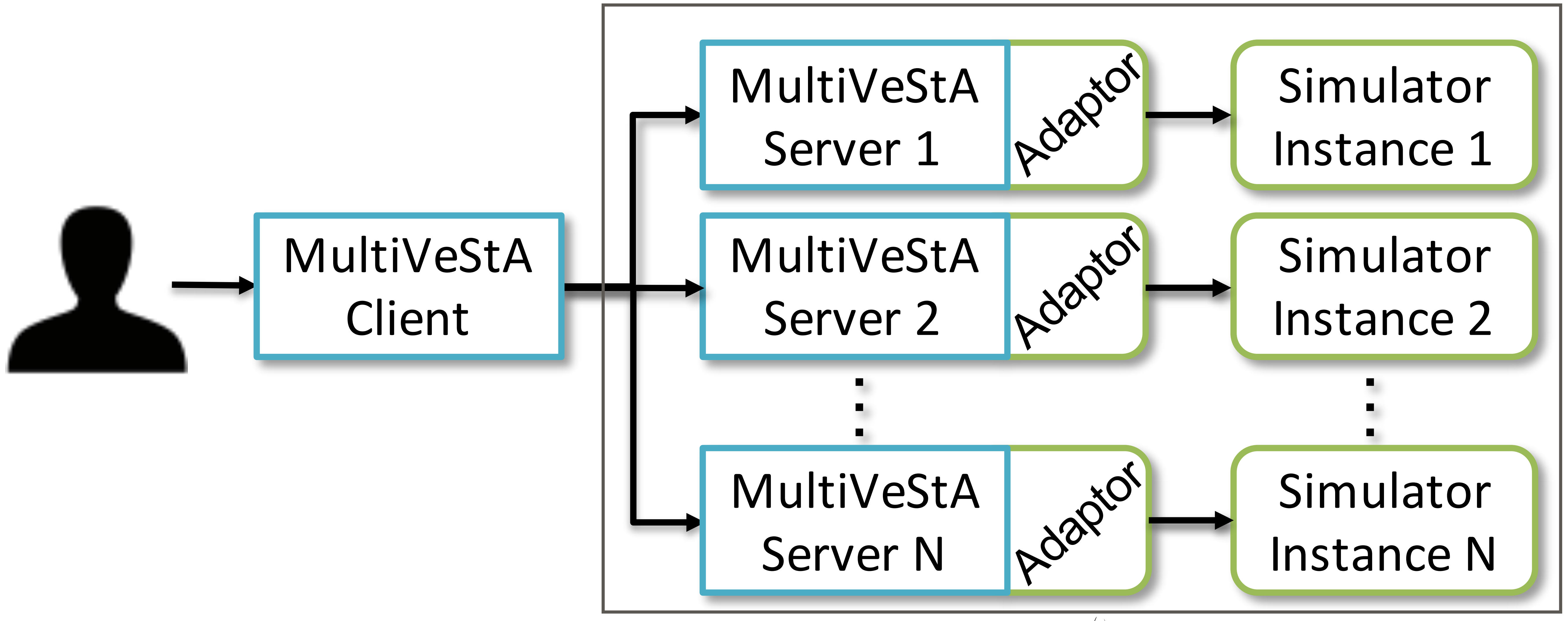
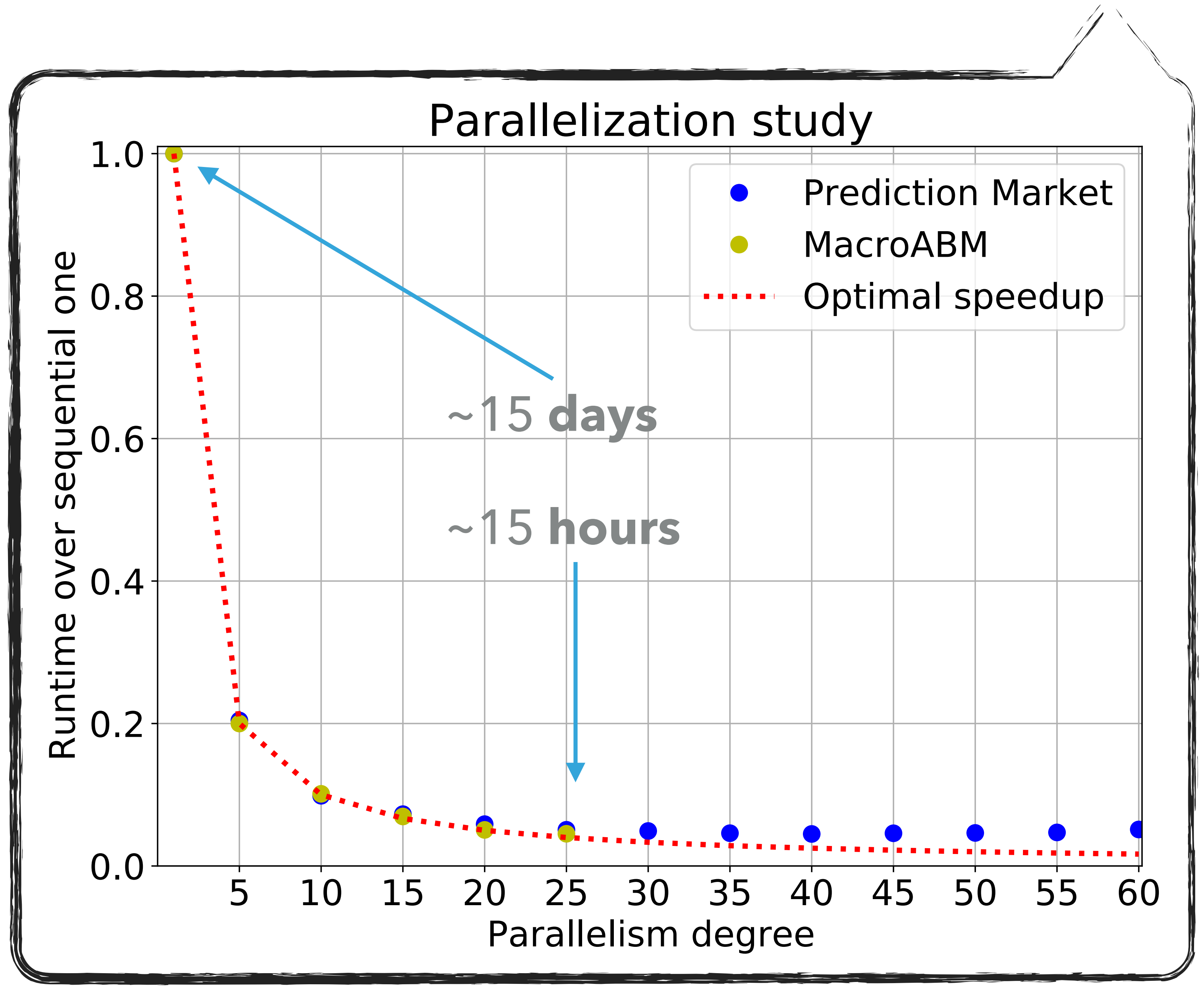




Java

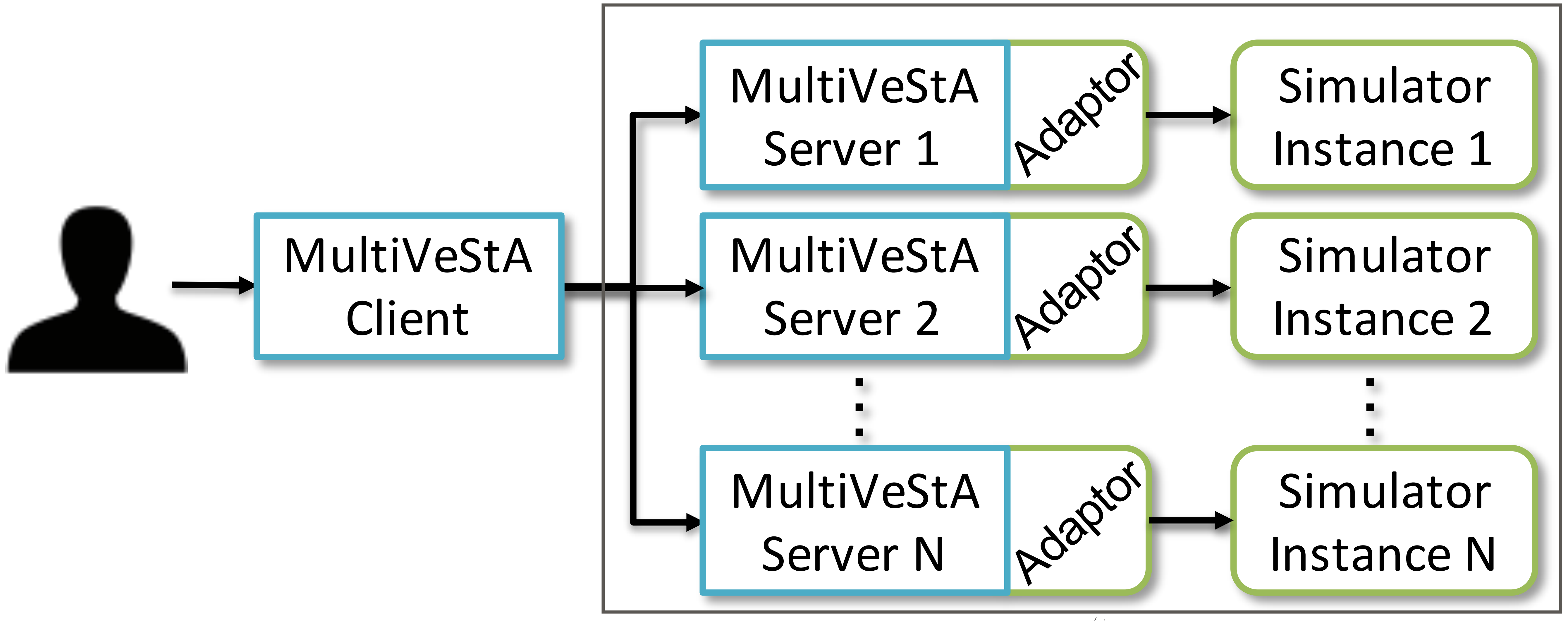
NetLogo

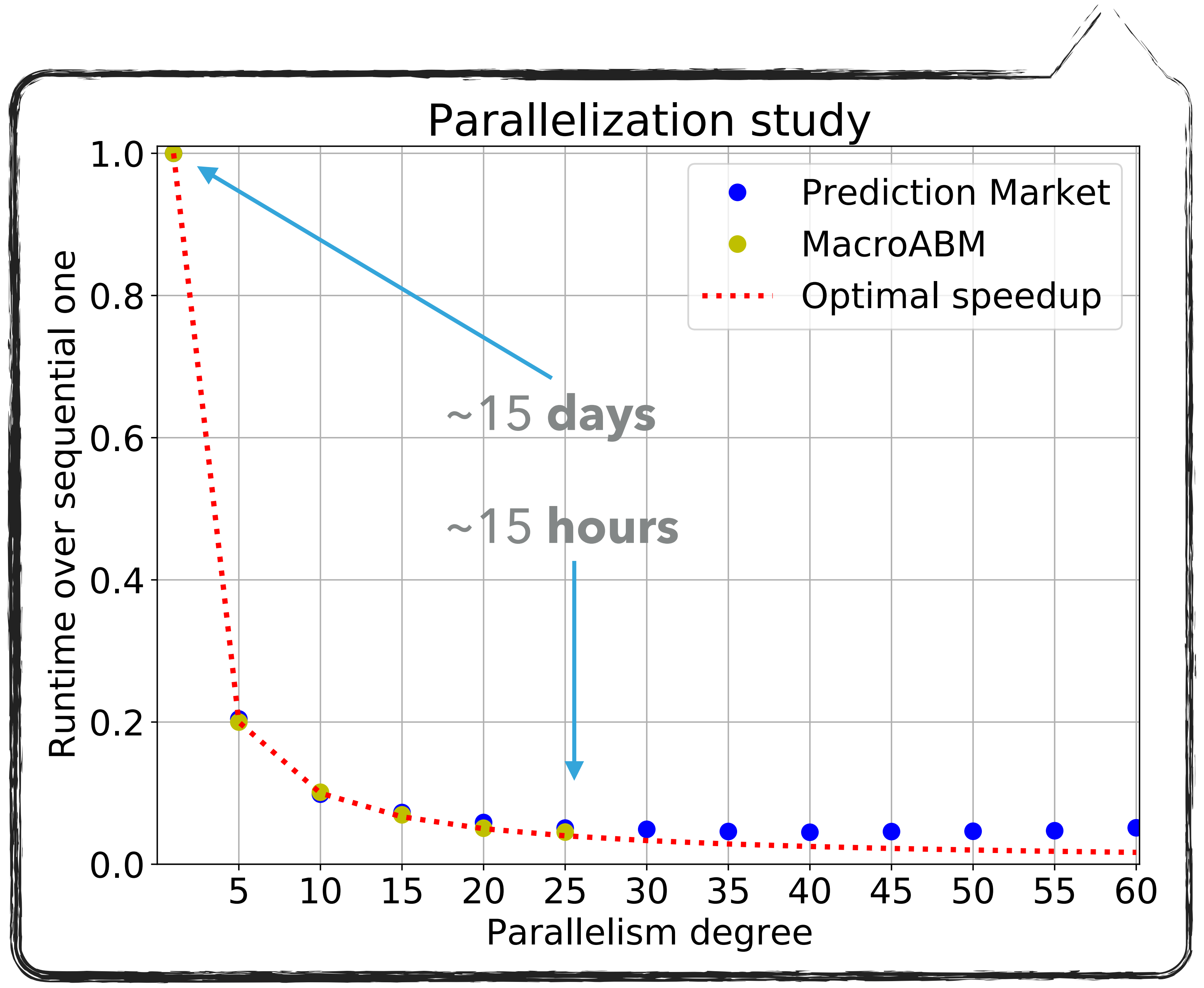
python

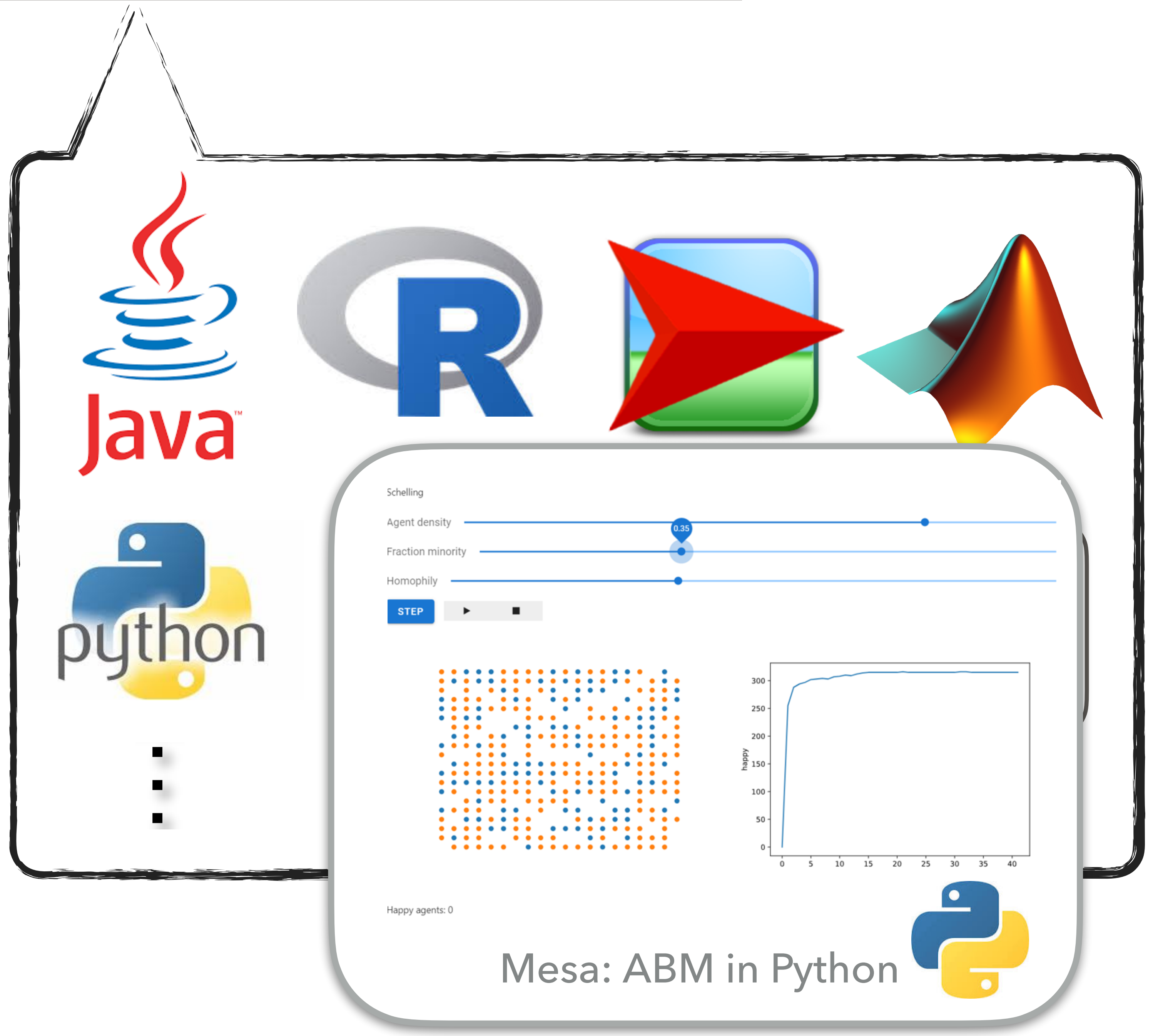
JMAB
JABM

MAUDE

MultiVeStA: SMC For Discrete-Event Simulators







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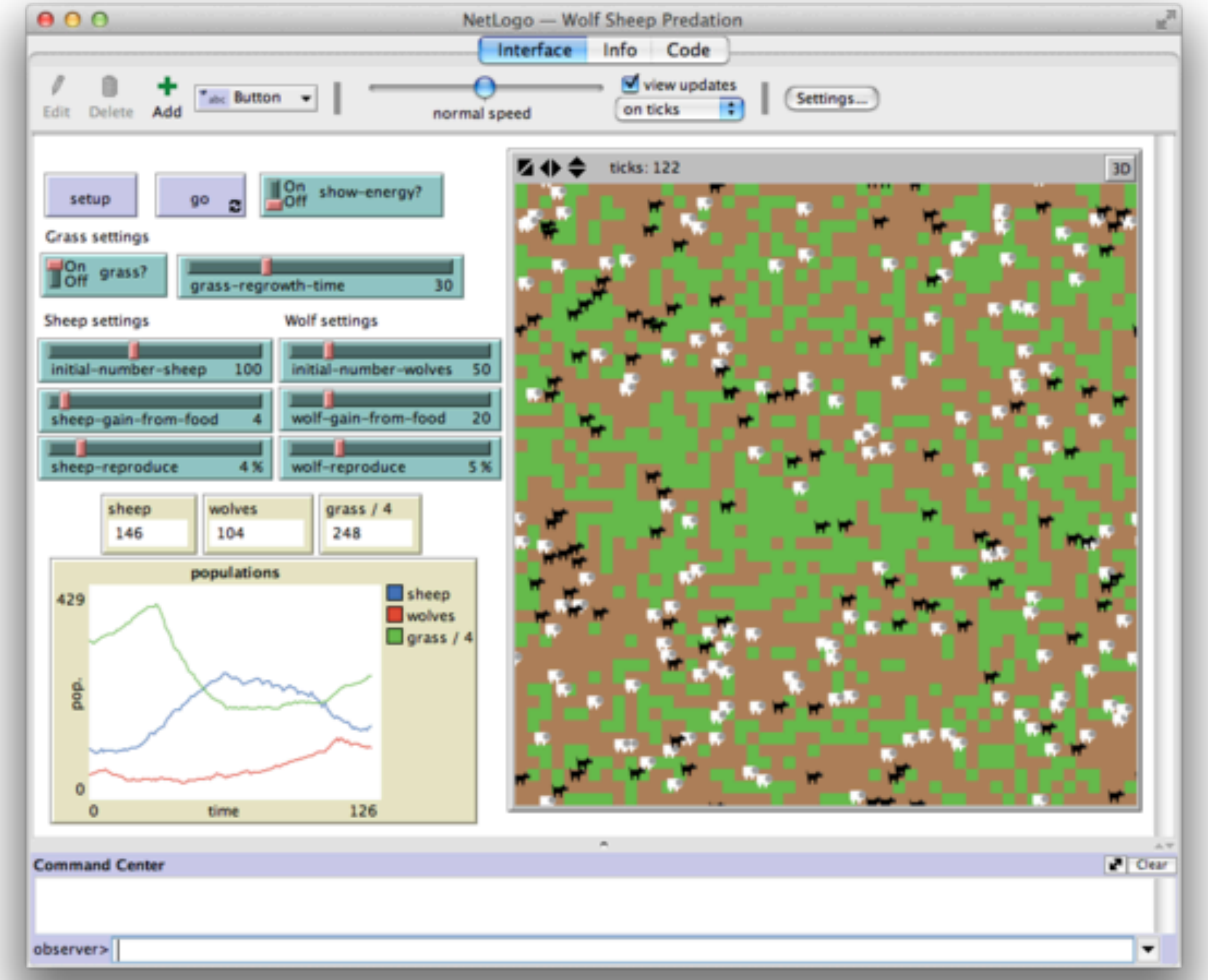
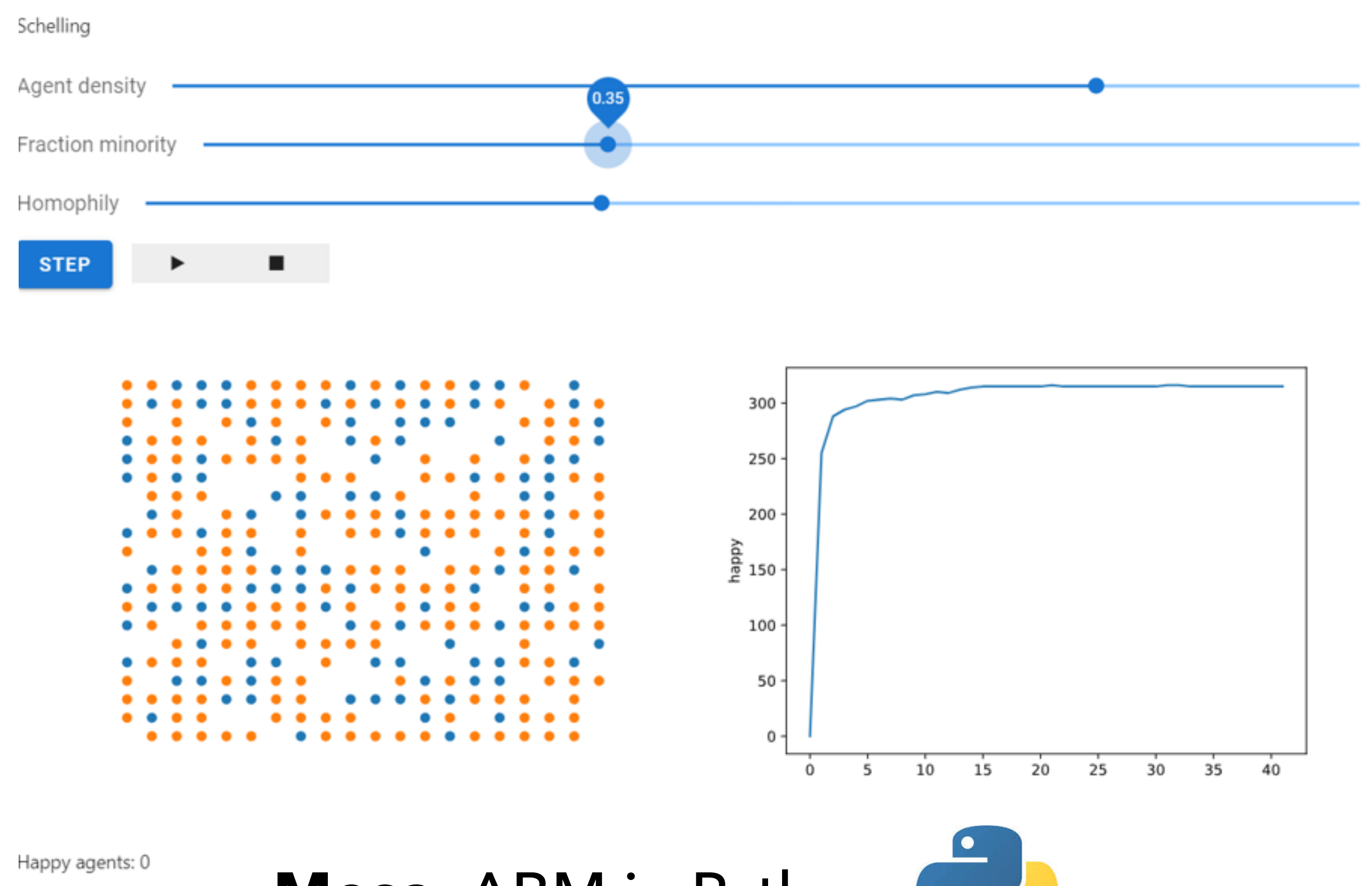
Two Popular Frameworks for ABM Modeling



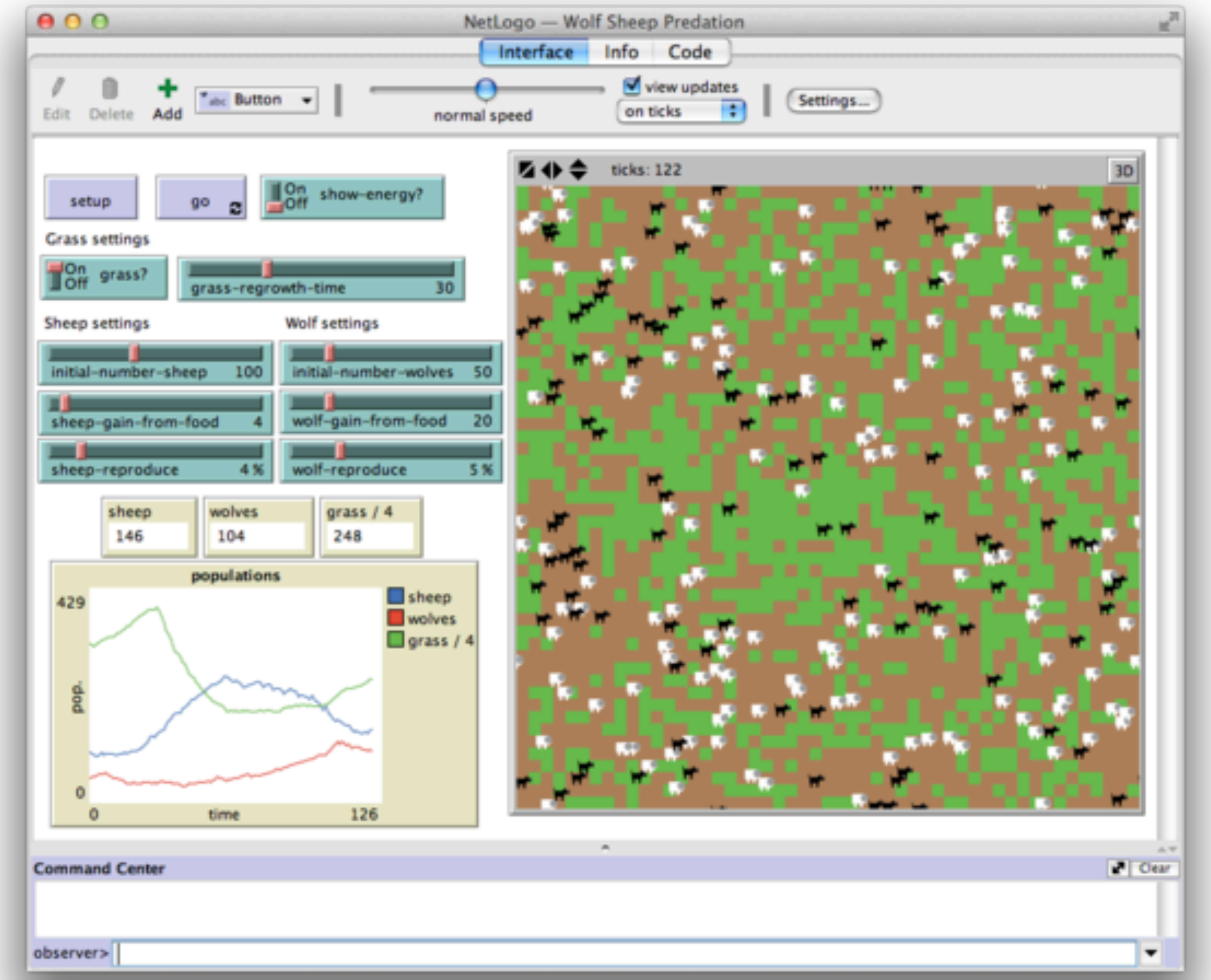
NetLogo multi-agent modeling
millions of students/teachers/researchers
https://ccl.northwestern.edu/netlogo/models/



Mesa: ABM in Python

From some points of view, a sort of 'porting' of netlogo in python…
https://github.com/projectmesa/mesa-examples

Two Popular Frameworks for ABM Modeling



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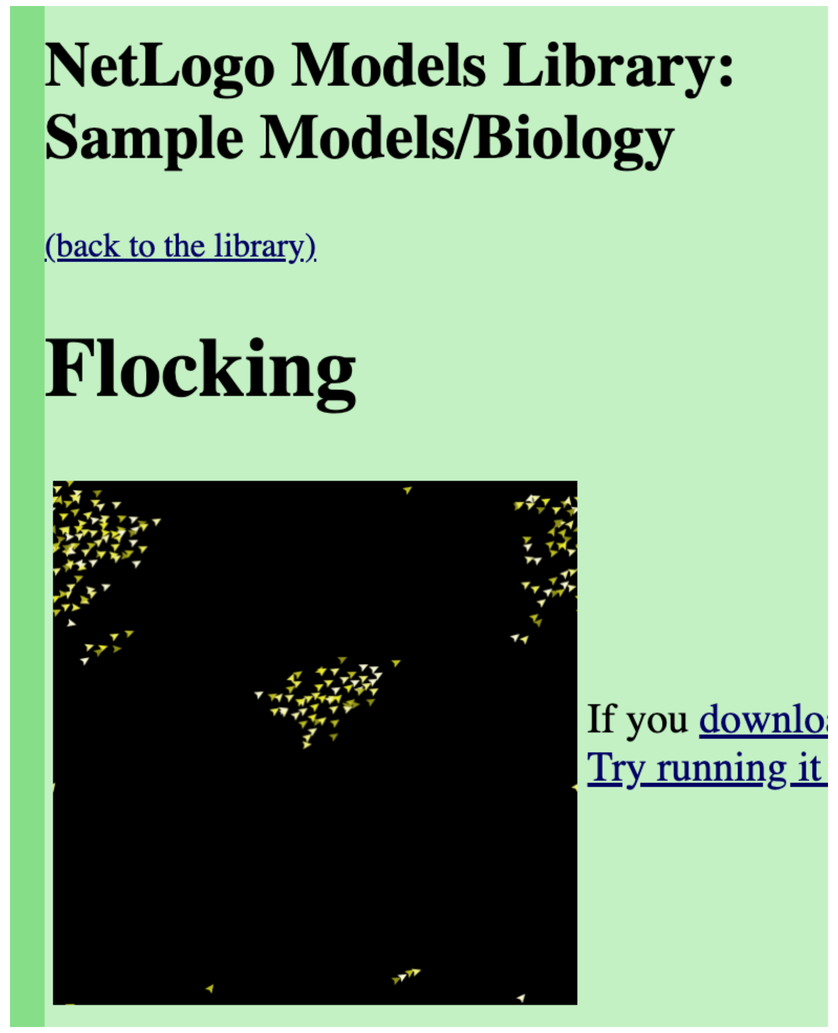

NetLogo Models Library:
Sample Models/Biology

(back to the library)

Flocking

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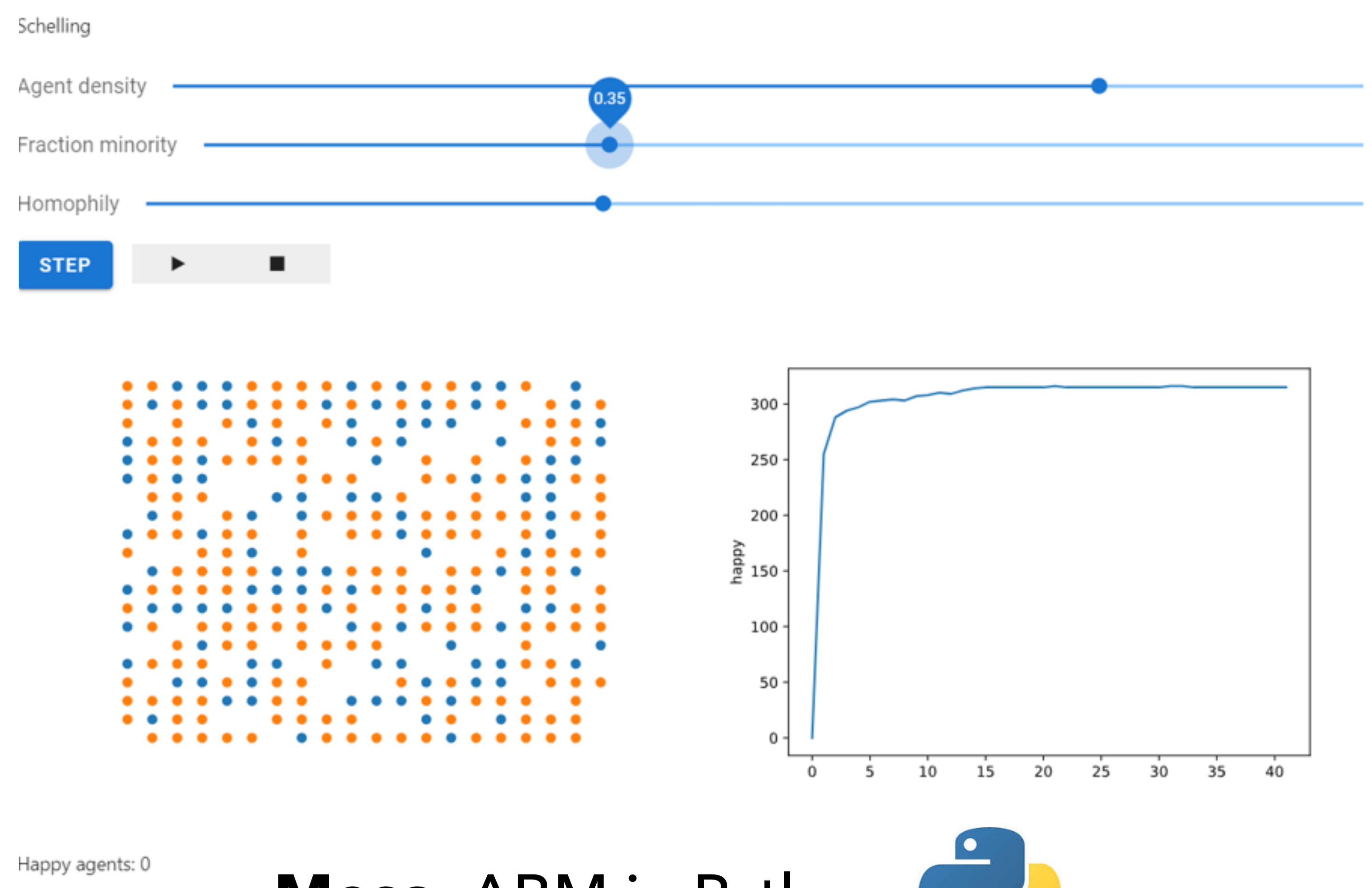
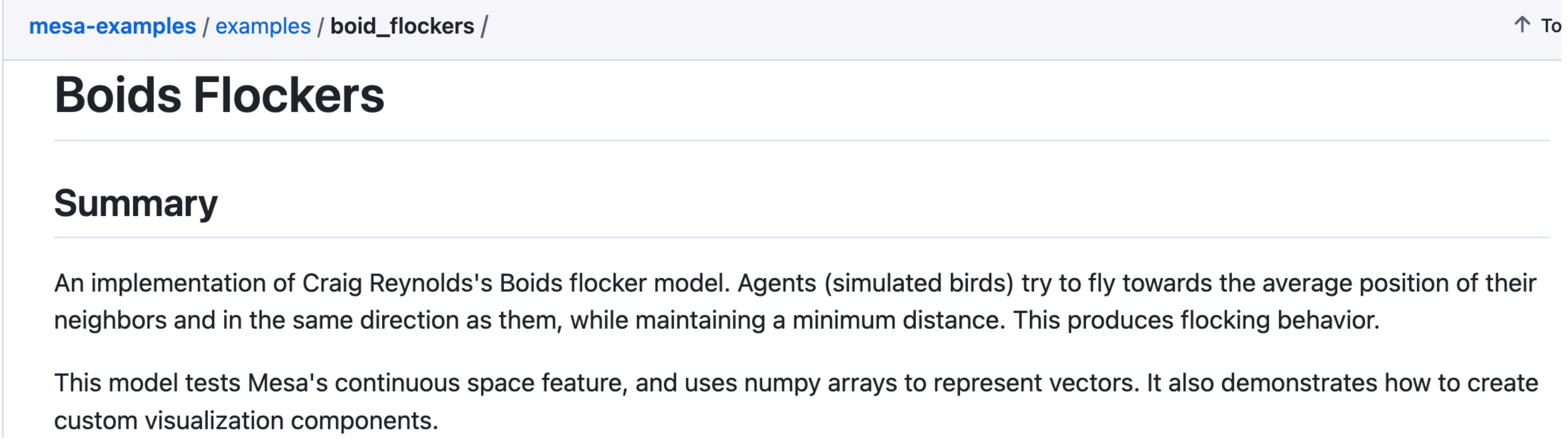


Mesa: ABM in Python

From some points of view, a sort of 'porting' of netlogo in python…
https://github.com/projectmesa/mesa-examples



mesa-examples / examples / boid_flockers /

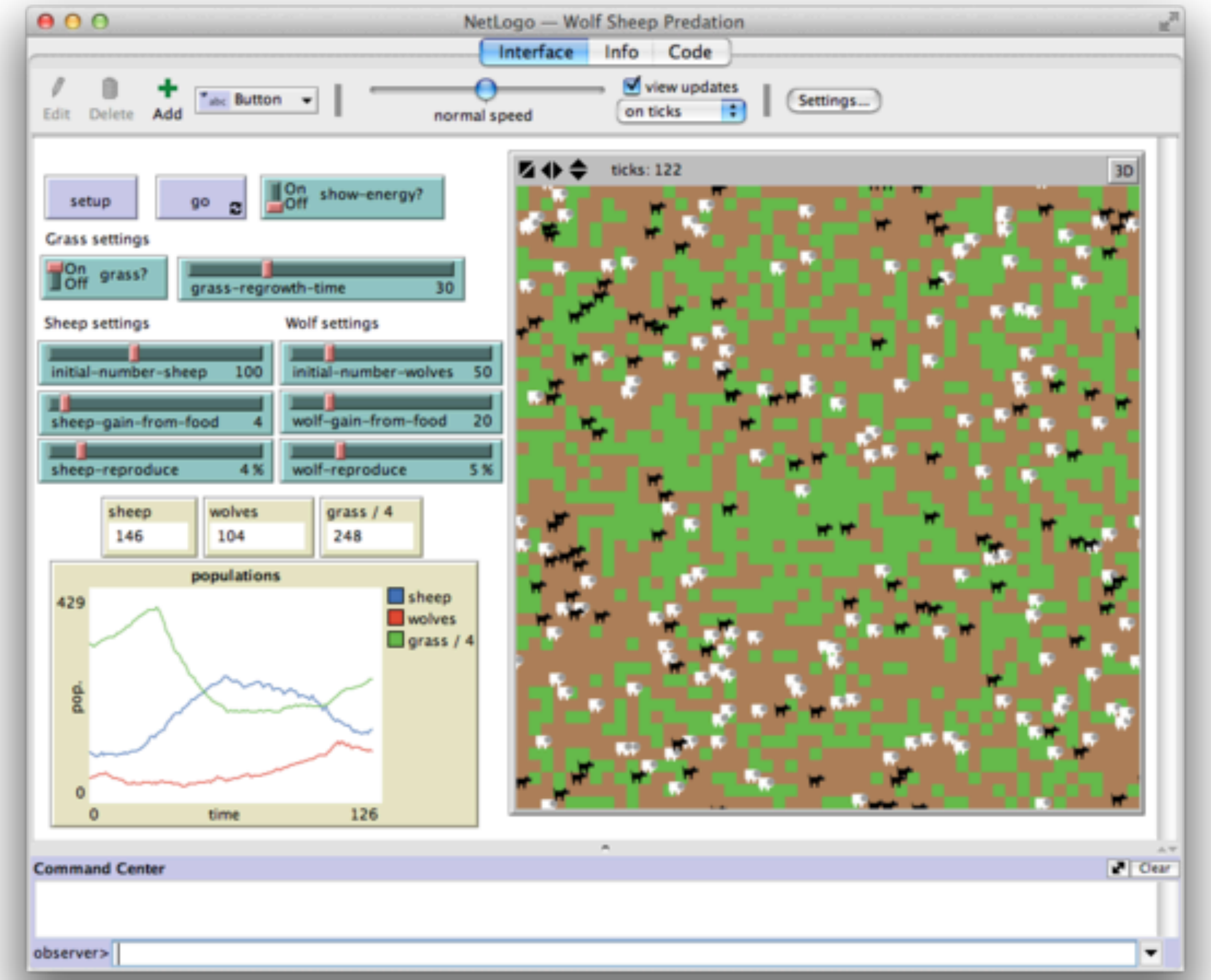
Boids Flockers

Summary

An implementation of Craig Reynolds's Boids flocker model. Agents (simulated birds) try to fly towards the average position of their neighbors and in the same direction as them, while maintaining a minimum distance. This produces flocking behavior.

This model tests Mesa's continuous space feature, and uses numpy arrays to represent vectors. It also demonstrates how to create custom visualization components.

Two Popular Frameworks for ABM Modeling



NetLogo multi-agent modeling
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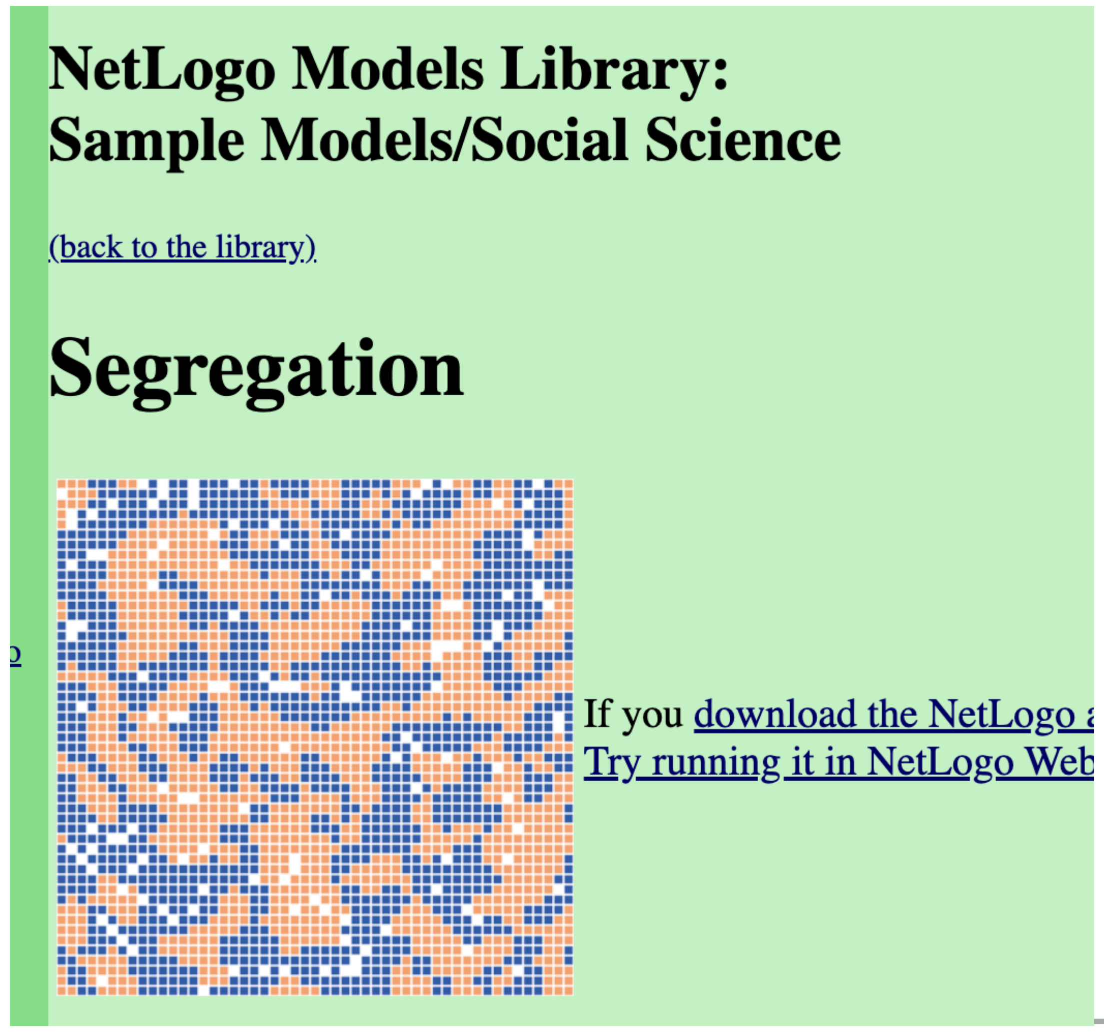

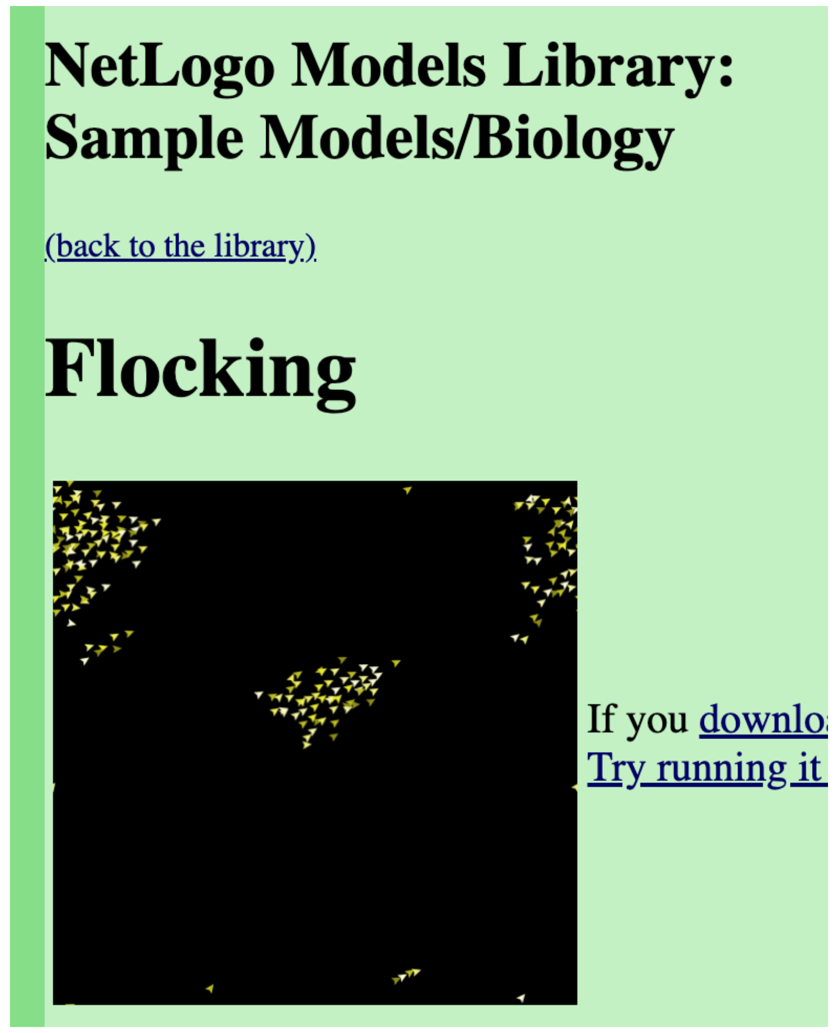

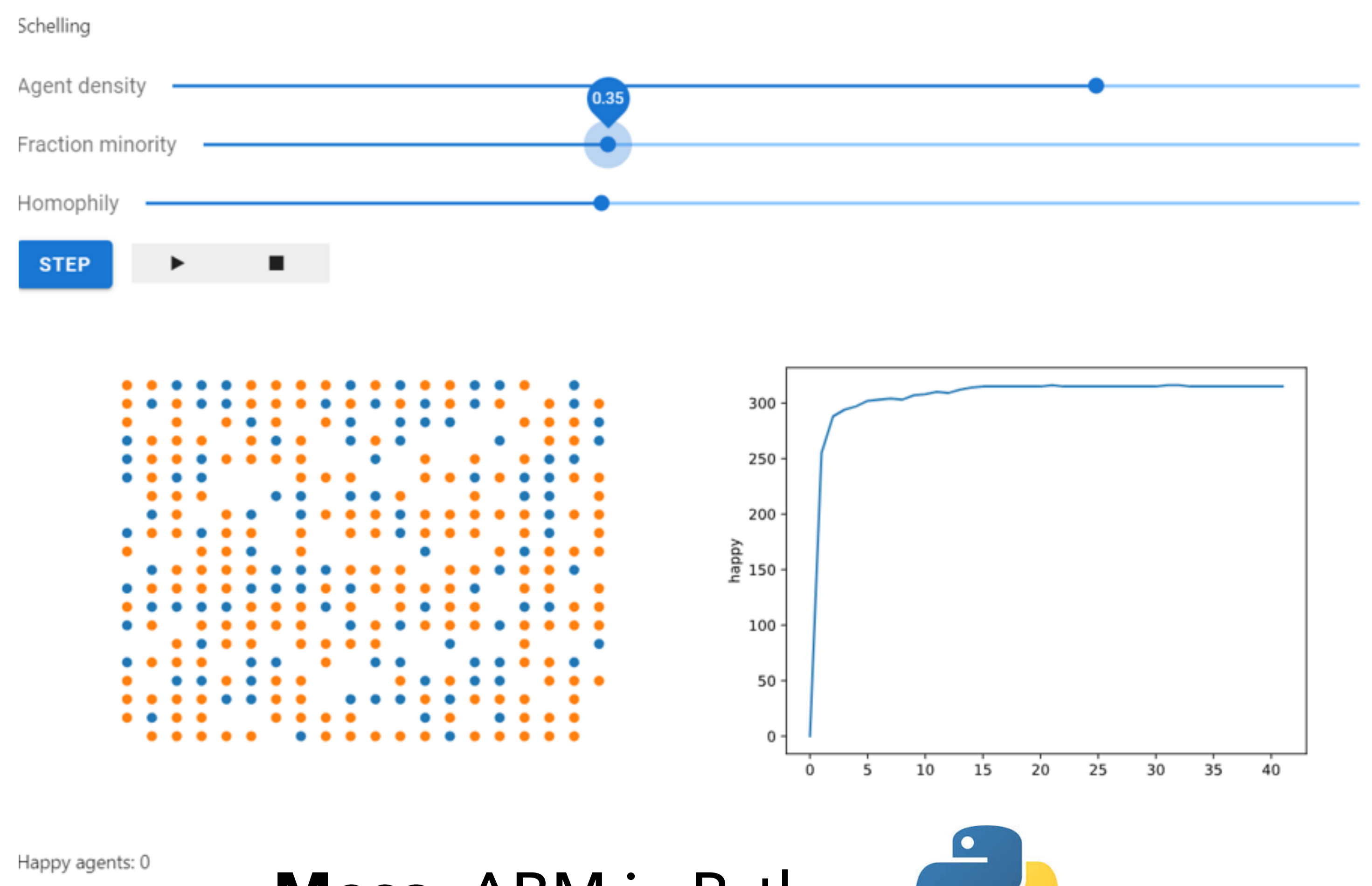

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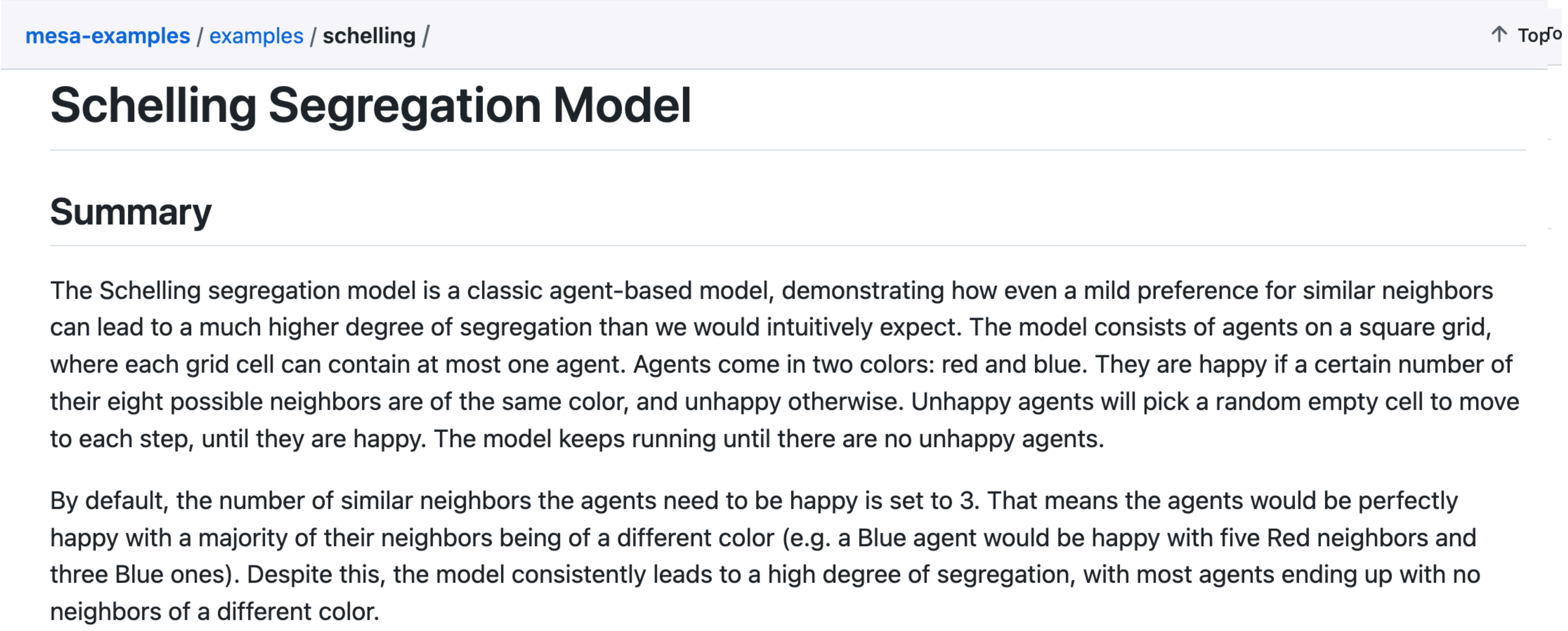

mesa-examples / examples / schelling /

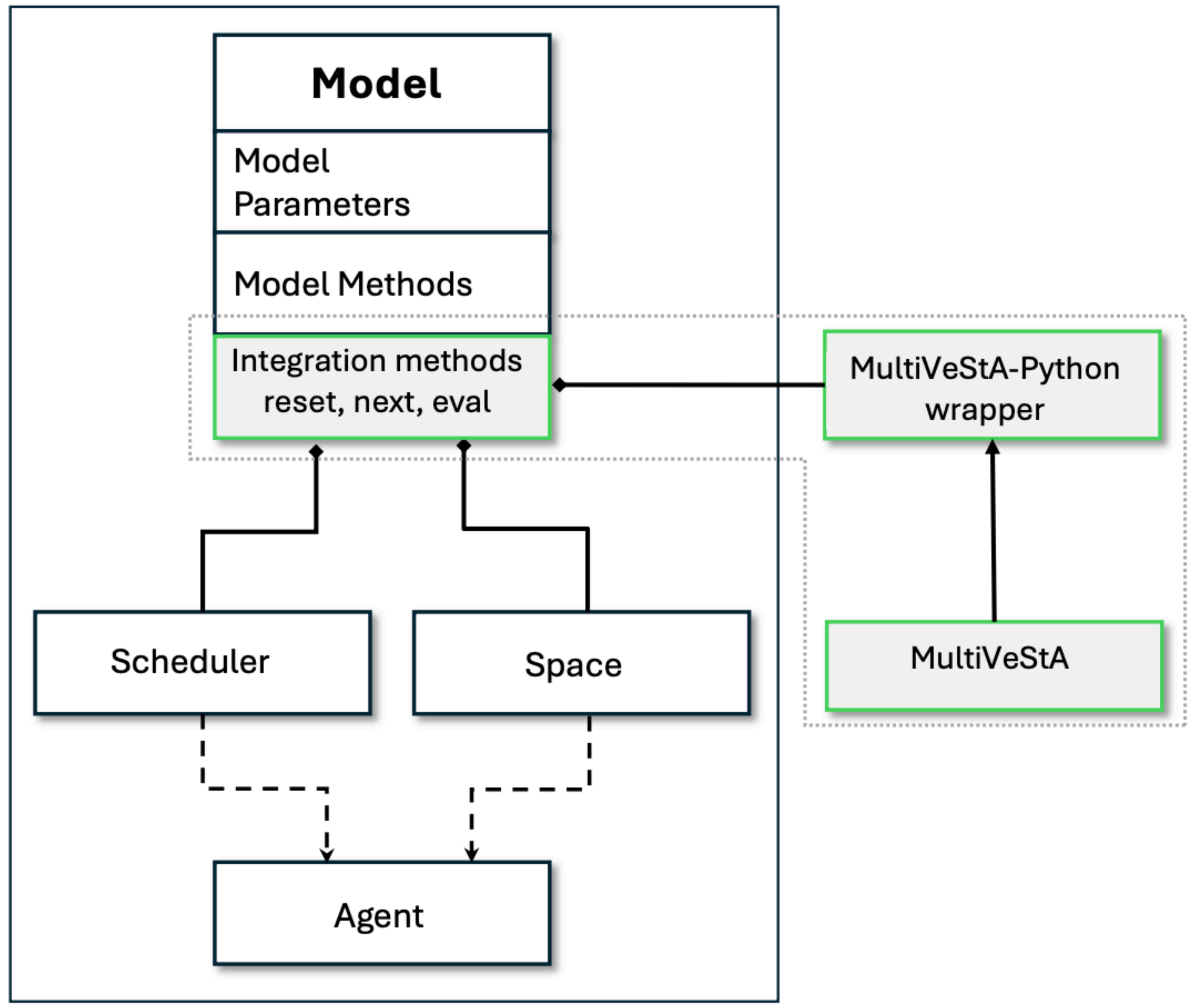
Schelling Segregation Model

Summary

The Schelling segregation model is a classic agent-based model, demonstrating how even a mild preference for similar neighbors can lead to a much higher degree of segregation than we would intuitively expect. The model consists of agents on a square grid, where each grid cell can contain at most one agent. Agents come in two colors: red and blue. They are happy if a certain number of their eight possible neighbors are of the same color, and unhappy otherwise. Unhappy agents will pick a random empty cell to move to each step, until they are happy. The model keeps running until there are no unhappy agents.

By default, the number of similar neighbors the agents need to be happy is set to 3. That means the agents would be perfectly happy with a majority of their neighbors being of a different color (e.g. a Blue agent would be happy with five Red neighbors and three Blue ones). Despite this, the model consistently leads to a high degree of segregation, with most agents ending up with no neighbors of a different color.

Integrating Mesa and MultiVeStA: Flocks of Boids




```python
class BoidFlockers(mesa.Model):
  def __init__(self, parameters):
    self.parameters = parameters
    #part of model initialization moved to set_simulator_for_new_simulation
    #self.make_agents()

  def make_agents(self):
    ...

  def step(self): #Method already implemented, no need to change
    self.schedule.step()

  #Code below is specific for integration with MultiVeStA

  def set_simulator_for_new_simulation(self,rnd_seed):
    random.seed(rnd_seed)
    self.random.seed(rnd_seed)
    self.reset_randomizer(rnd_seed)
    #part of model initialization moved here from the constructor
    self.make_agents()
    ...

  def eval(self, obs):
    if obs == 'avg_distance_from_centroid':
      ...
    elif obs == 'avg_visible_neighbors':
      ...
```

Listing 1: Code snippet of Mesa's Boids flockers integrated with MultiVeStA

Integrating Mesa and MultiVeStA: Flocks of Boids

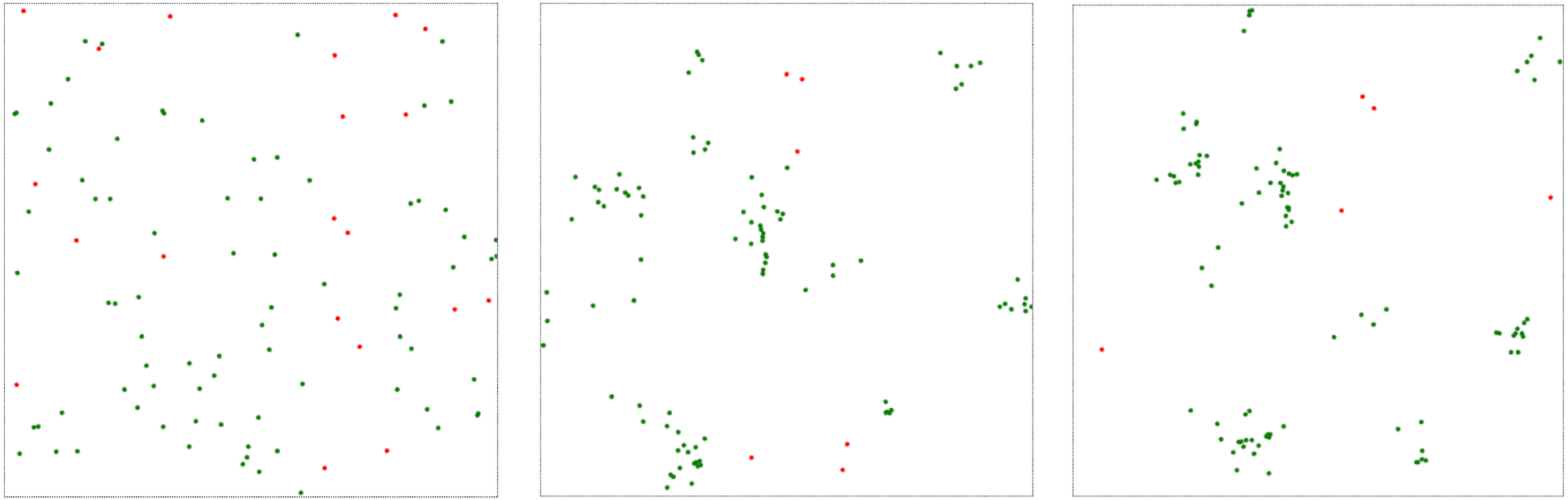


Fig. 3: Two snapshots from Mesa GUI for a simulation of the Boids model.

Integrating Mesa and MultiVeStA: Flocks of Boids

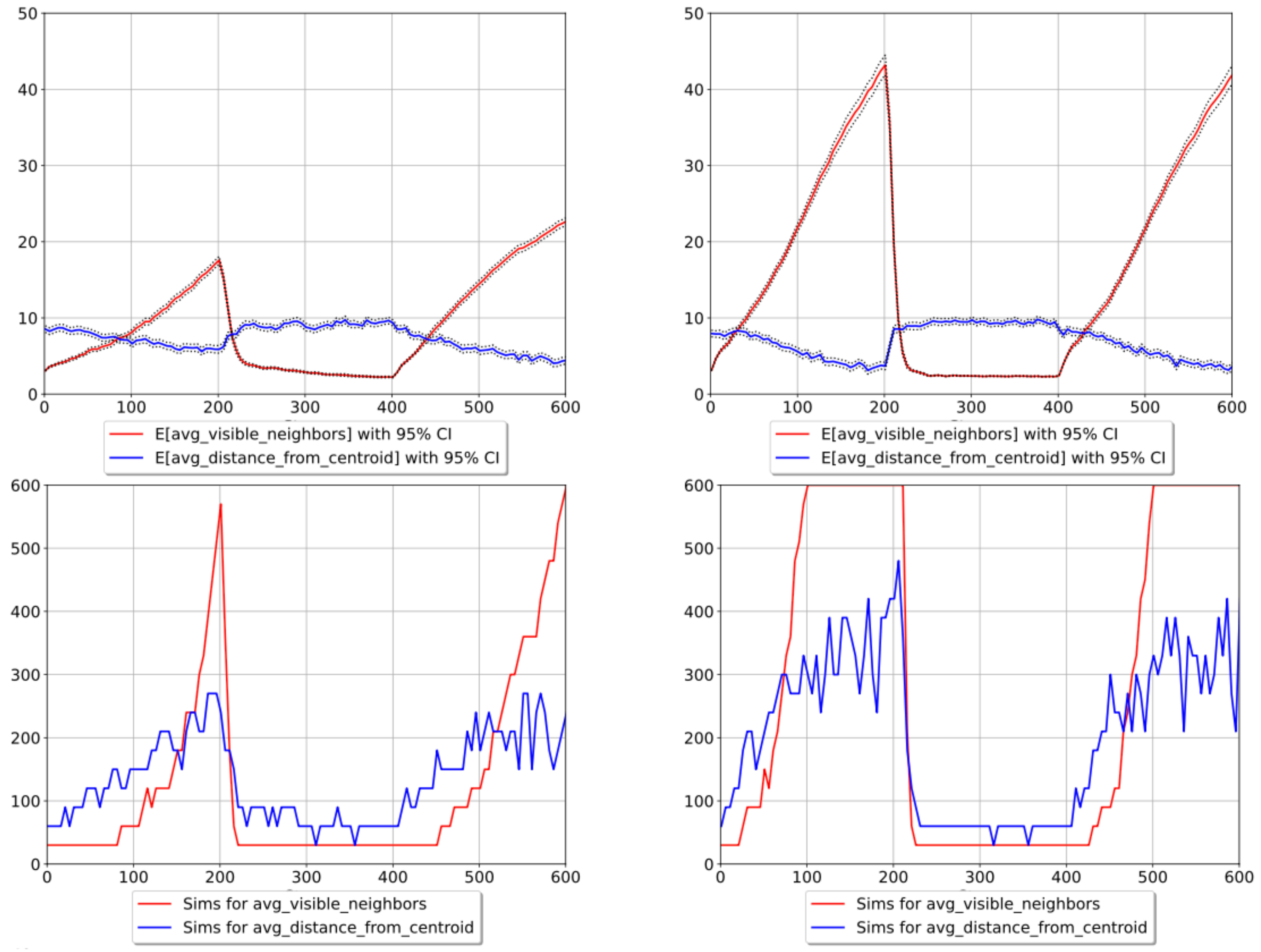


Fig. 4: MultiVeStA results for Listing 2. The predator arrives at step 200, and its effect ends at step 400. Left: results, number of simulations, and sample variance for the default cohesion weight 0.03. Right: same as for cohesion weight 0.09.

```
ObsAtStep(step,obs) =
        if ( s.rval("steps") == step )
                then s.rval(obs)
                else next(ObsAtStep(step,obs)) fi ;
eval parametric(E[ ObsAtStep(x,"avg_distance_from_centroid")],
                    E[  ObsAtStep (x,"avg_visible_neighbors")],x,1,5,601) ;
```

Listing 2: MultiQuaTEx query for the Boids flocker model

Integrating Mesa and MultiVeStA: Schelling Segregarion Model

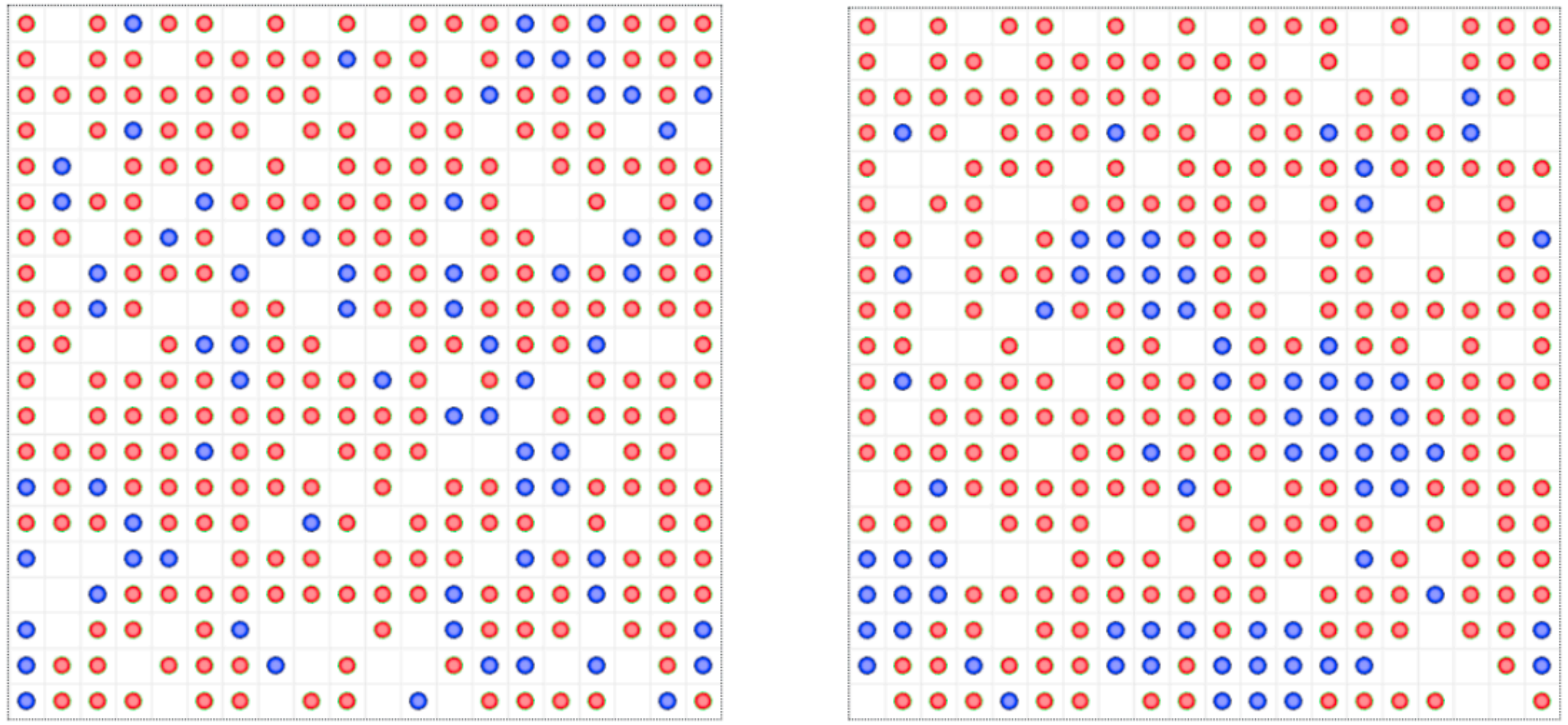


Fig. 7: Two snapshots from Mesa GUI for a simulation of the Schelling model.

Integrating Mesa and MultiVeStA: Schelling Segregarion Model

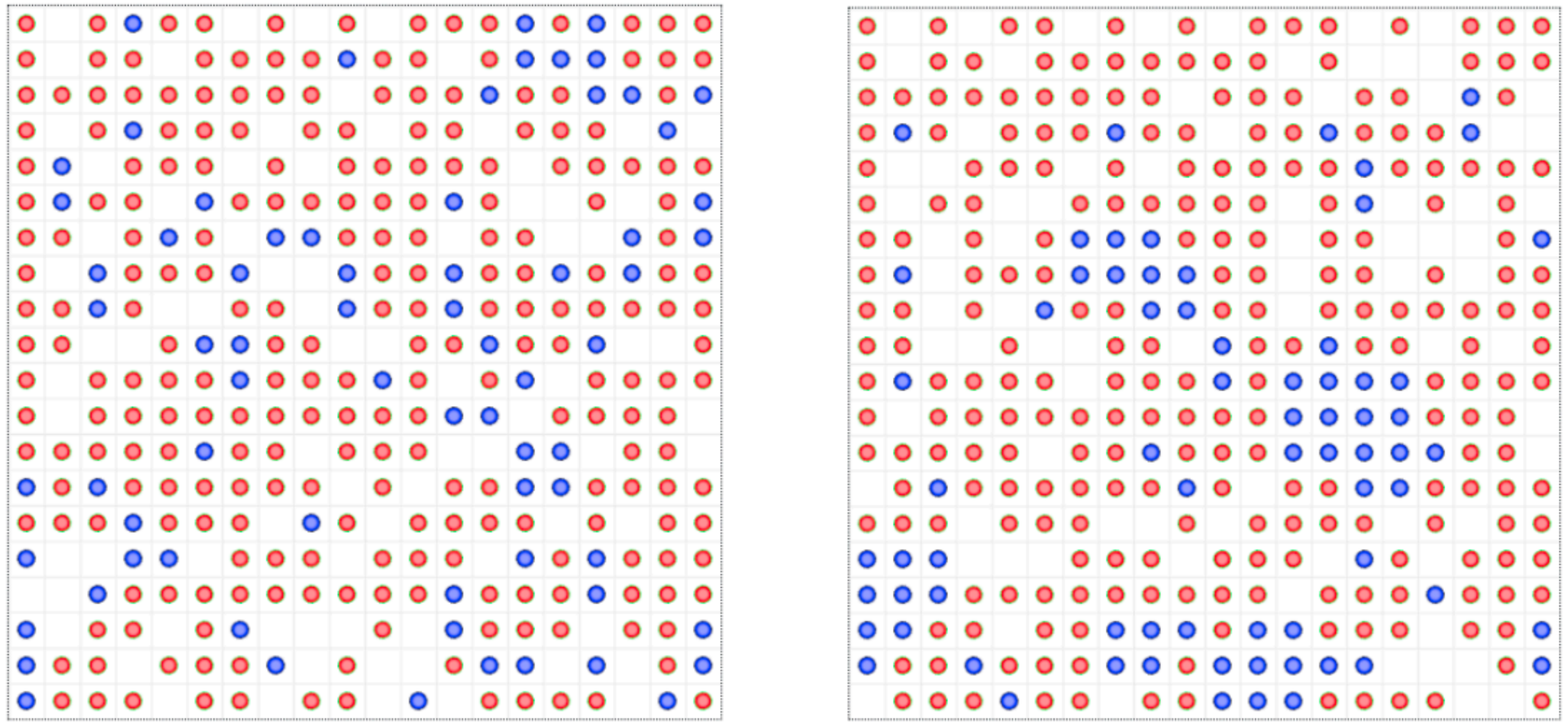


Fig. 7: Two snapshots from Mesa GUI for a simulation of the Schelling model.

```
Obs(obs) = s.rval(obs);
eval autoRD(E[ Obs("ratioHappy")]);

Obs(obs) = s.rval(obs);
eval autoBM(E[ Obs("ratioHappy")]);
```

Listing 3: MultiQuaTEx queries for the Schelling model

Asking for: 95% CI with width at most 0.1

AutoRD: Warmup ends at 1024 steps. Afterwards, 30 simulations up to 2048 steps are enough. We get estimate: 0.98

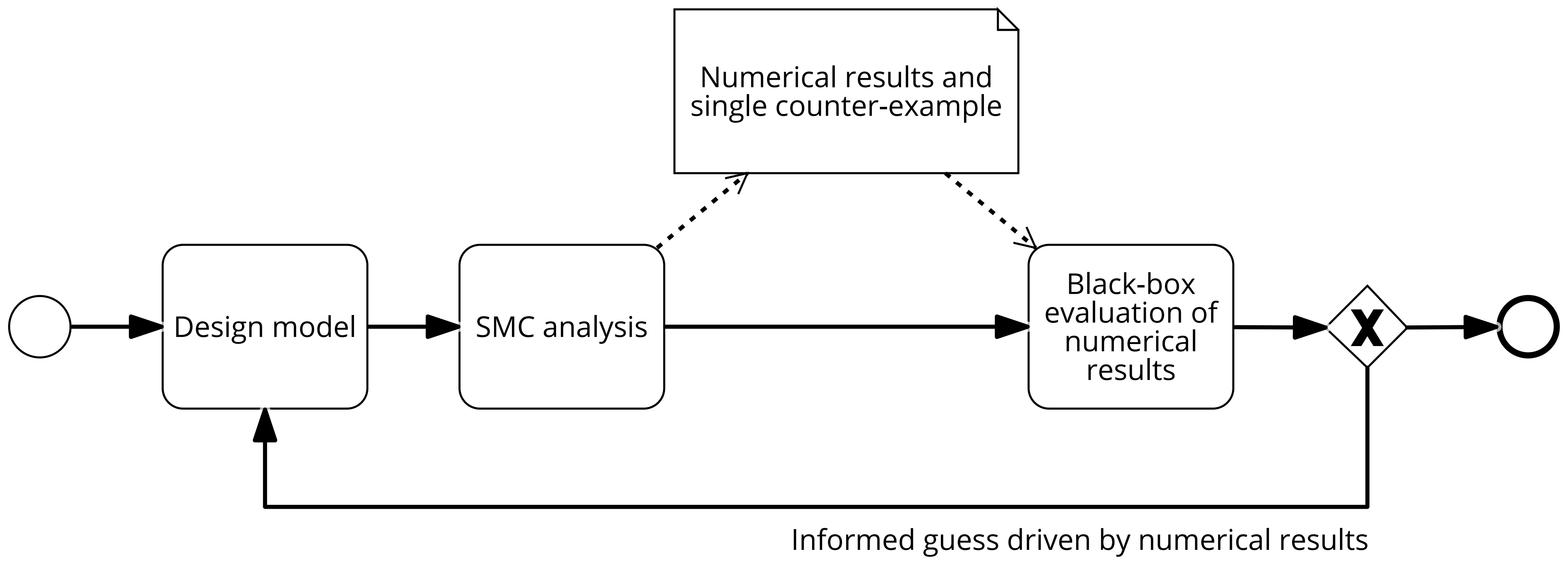
AutoBM: Warmup ends at 1024 steps. Afterwards, we had to get to 4096 steps. We get estimate: 1.02

Ergodicity diagnostics: No deviation observed

Outline

- The context at Sant’Anna Pisa
- Economic ABMs
- SMC of Economic ABMs & MultiVeStA
- Two frameworks for ABMs from the social sciences: NetLogo and Mesa
  - Mesa case study: Transient analysis of flocks of Boids
  - Mesa case study: Steady state analysis of Segregation models (Schelling)
- Recent results on graphically explaining SMC results

Can we explain the results of SMC?



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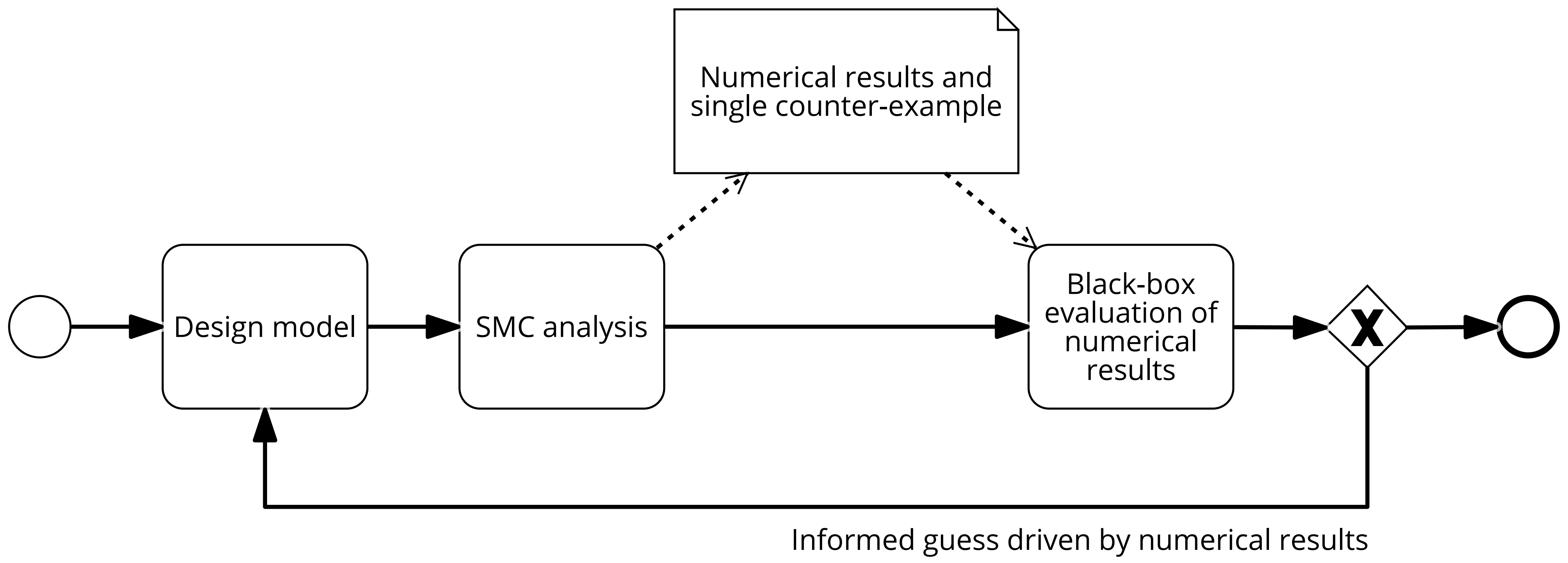

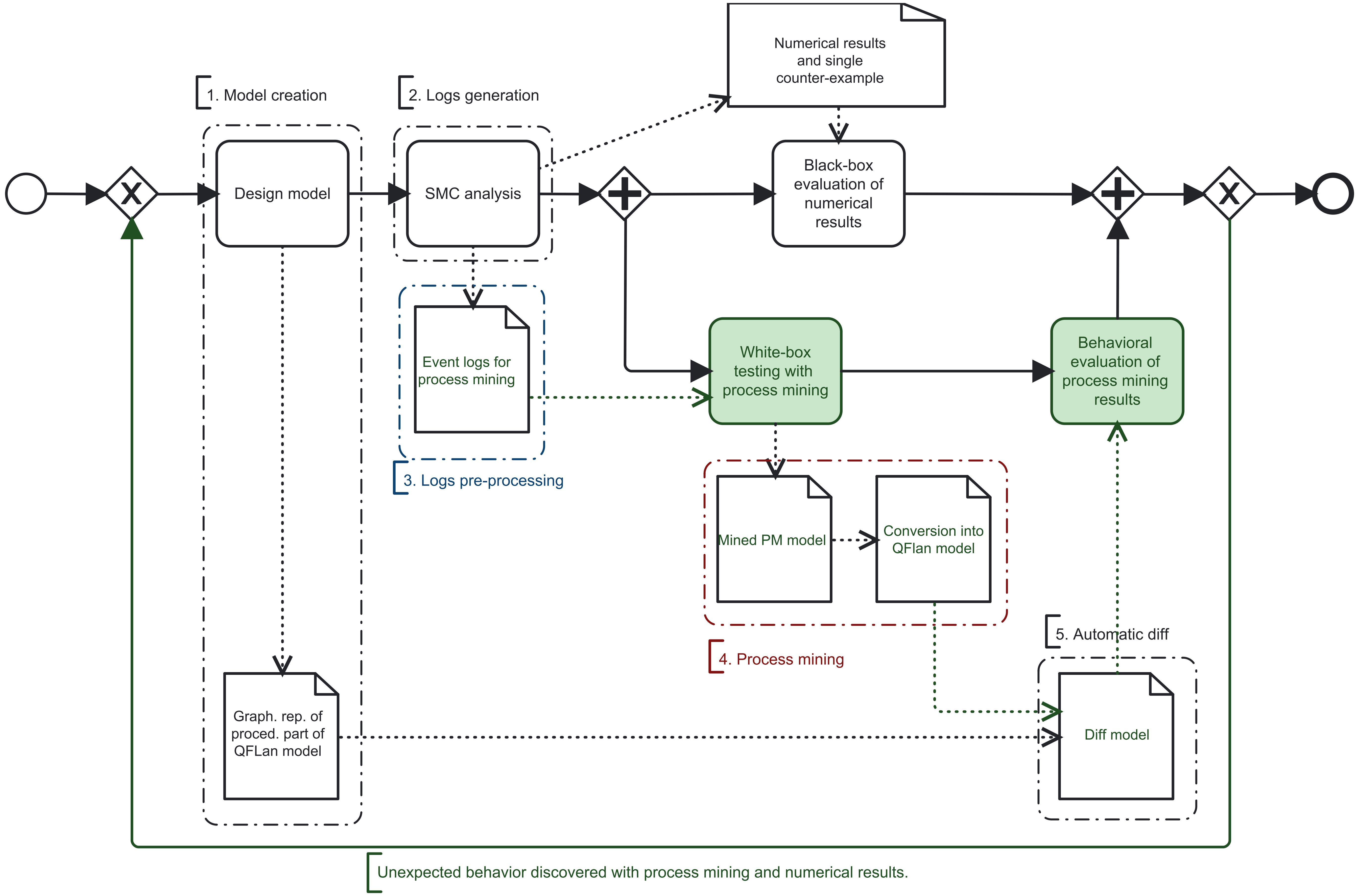

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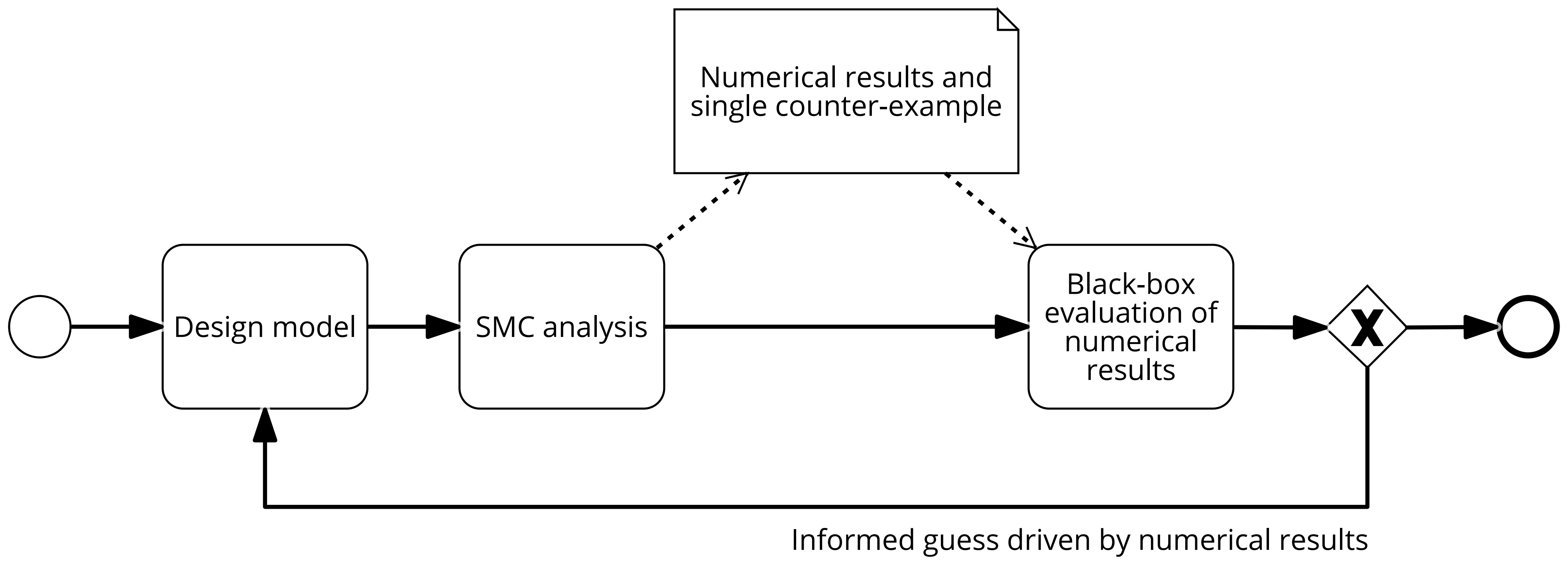

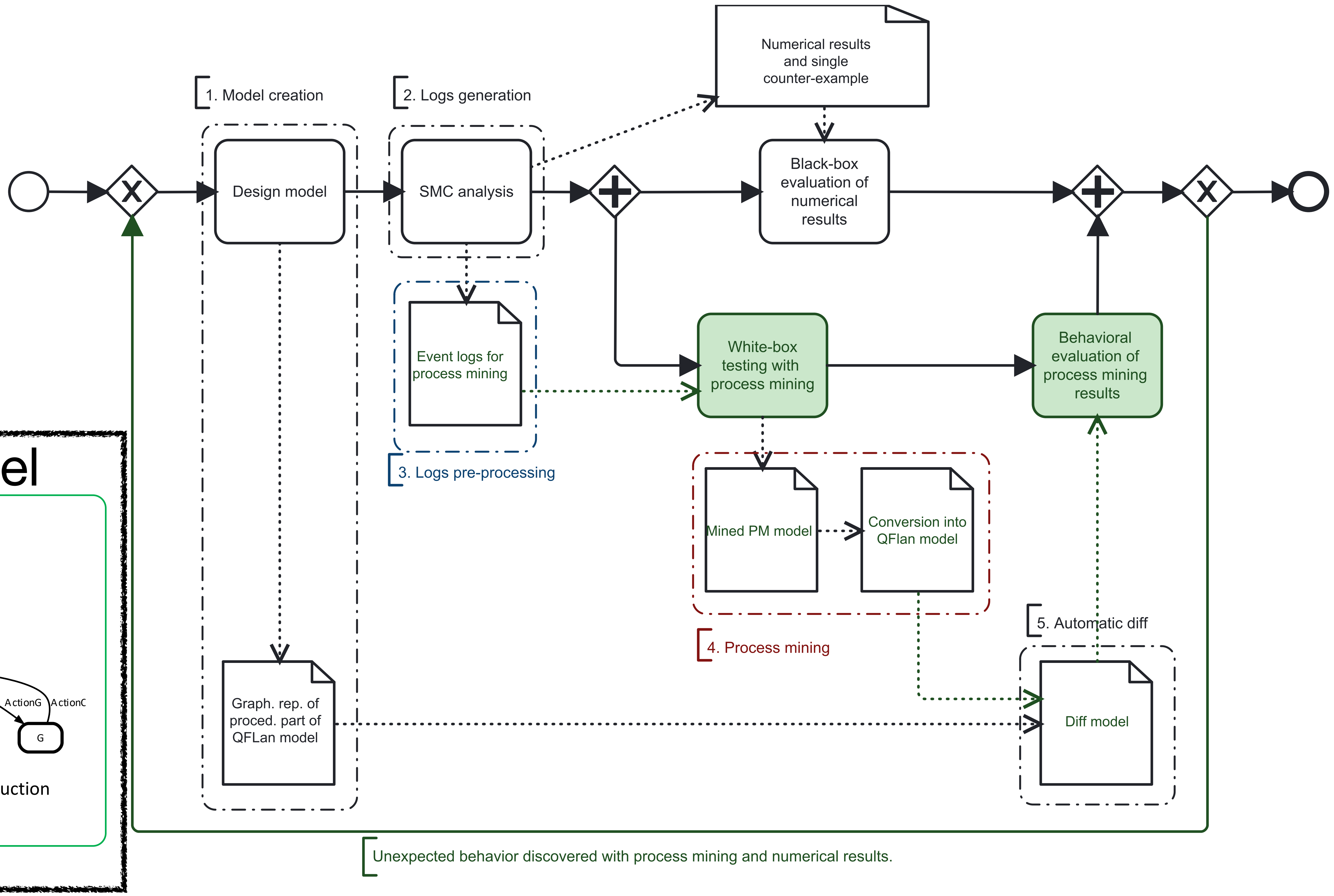

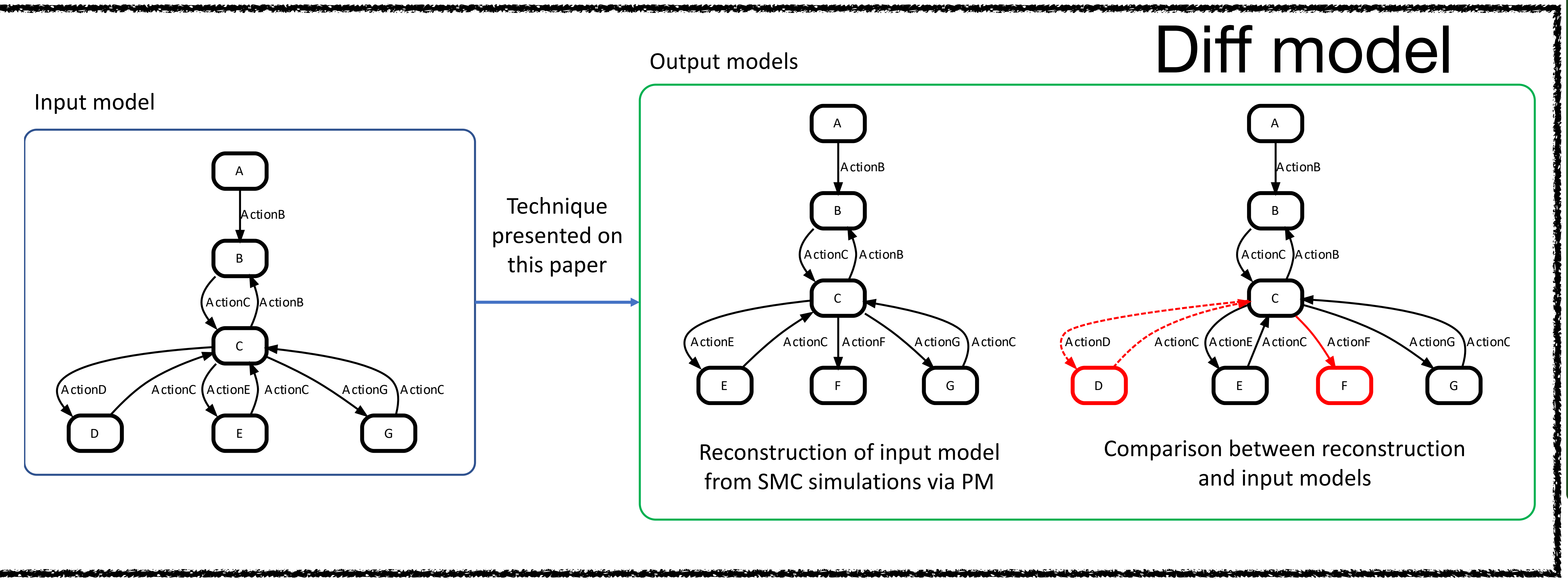

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THANK YOU FOR YOUR ATTENTION!

QUESTIONS? FEEDBACK?

JEDC Paper

https://www.sciencedirect.com/science/article/abs/pii/S0165188922001634

Tool and models available at:

https://bit.ly/MultiVeStATool

https://github.com/andrea-vandin/MultiVeStA/wiki/Integration-with-Mesa

This work has been partially supported by the project: SMaRT COnSTRUCT:
Statistical Methods & Reduction Techniques for learning from COmplex, STRUCTured data,
CUP J53C24001460006, funded in the context of the programma di Ricerca e Innovazione FAIR "Future Artificial Intelligence Research" Code PE0000013

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Statistical Model Checking of Black-Box Probabilistic Systems

CAV 2004

Koushik Sen, Mahesh Viswanathan, and Gul Agha

Department of Computer Science
University of Illinois at Urbana-Champaign
{ksen,vmahesh,agha}@uiuc.edu

On Statistical Model Checking of Stochastic Systems

CAV 2005 VESTA

Koushik Sen, Mahesh Viswanathan, and Gul Agha

Department of Computer Science,
University of Illinois at Urbana-Champaign
{ksen, vmahesh, agha}@uiuc.edu

Probabilistic Verification for "Black-Box" Systems*

CAV 2005

Håkan L.S. Younes

Computer Science Department, Carnegie Mellon University,
Pittsburgh, PA 15213, USA

PVeStA: A Parallel Statistical Model Checking and Quantitative Analysis Tool

CALCO 2011 PVESTA

Musab AlTurki and José Meseguer

University of Illinois at Urbana-Champaign,
Urbana, IL 61801, USA

MultiVeStA: Statistical Model Checking for Discrete Event Simulators *

VALUETOOLS 2013

Stefano Sebastio
IMT Institute for Advanced Studies Lucca, Italy
stefano.sebastio@imtlucca.it

Andrea Vandin
ECS, University of Southampton, UK,
IMT Institute for Advanced Studies Lucca, Italy
a.vandin@soton.ac.uk

Automated and Distributed Statistical Analysis of Economic Agent-Based Models

Journal of Economic Dynamics and Control, 2022

Andrea Vandin[a,b], Daniele Giachini[a], Francesco Lamperti[a,d], Francesca Chiaromonte[a,c]

White-box validation of quantitative product lines by statistical model checking and process mining

Journal of Systems and Software, 2024

Roberto Casaluce[a], Andrea Burattin[b], Francesca Chiaromonte[a,c], Alberto Lluch Lafuente[b], Andrea Vandin[a,b,*]

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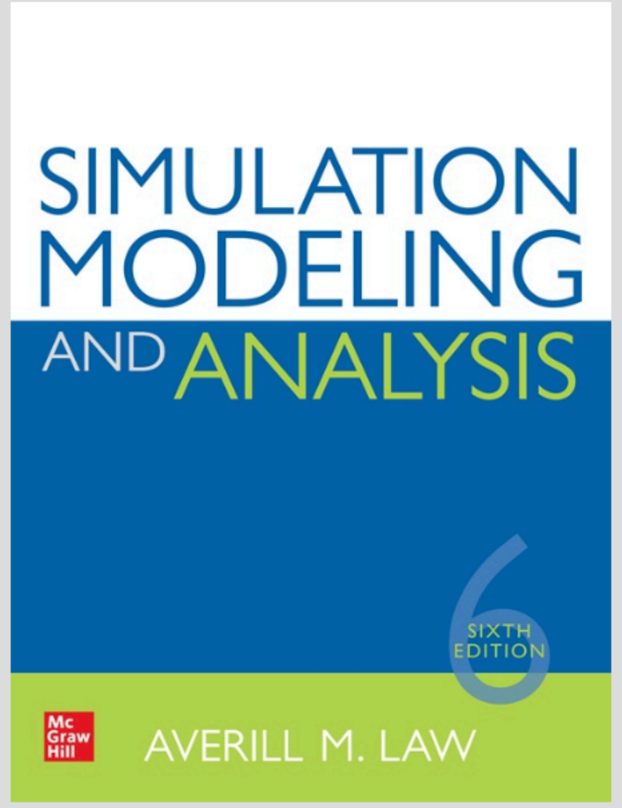

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McGraw-Hill, 2024, 688 pages